An Incestuous Community

How Salt, Tires, and Explosives Built America's Strategic Missiles

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Preface

Preface

This book began as a question: how did the American solid rocket motor industry — one of the most technically specialized, capital-intensive, and strategically consequential industrial sectors ever created — end up in the hands of a salt company, a tire manufacturer, a jar maker, and a thermostat conglomerate?

The answer turns out to be a history of industrial genealogy: of companies that split apart and recombined, of engineers who outlasted every corporate owner they worked for, and of propellant chemistry and design knowledge that transferred through people rather than through acquisition agreements.

The phrase *incestuous community* appears in oral histories of the American solid rocket industry so often that it almost becomes invisible. Engineers say it about colleagues who left Thiokol for Hercules and came back. Program managers say it about competitors who share the same propellant formulations derived from the same government research. Historians say it about corporate families where the same assets traded between the same holding companies across three decades.

This book traces the lineages. It is organized not by chronology but by bloodline — by corporate ancestor. Each chapter follows a single genealogical thread from its origin to its present form, then steps back to show how the threads braided together.

1.1 What This Book Is

A corporate and technical history of the American solid rocket motor industry from approximately 1912 to the present. The focus is on solid propulsion for strategic systems — ballistic missiles and their boosters — with secondary attention to space launch applications where they directly follow from the military heritage.

This is not a hagiography of engineers or a polemic about the defense industry. It is an attempt to understand how a technically coherent community of practice survived, through every corporate reorganization, the continuous institutional memory that the weapons required.

1.2 What This Book Is Not

This is not a technical handbook on solid propellant chemistry, internal ballistics, or motor design. Some technical context appears where it illuminates the business or historical narrative — why a particular formulation mattered, what made case design difficult — but readers seeking engineering depth should consult the technical literature.

It is also not a comprehensive catalog of every solid rocket program. Programs appear when they shaped the corporate genealogy or when their history illustrates the broader argument.

1.3 How to Read It

The chapters are organized by lineage. You may read them in order — the sequence follows a rough chronological logic within each corporate family — or you may follow a single thread from introduction to conclusion. The final two chapters synthesize the consolidation story and its implications.

A reader with deep familiarity with one lineage (say, Thiokol from the Challenger inquiry) will find the corporate context of adjacent lineages illuminating. A reader new to the industry may find it useful to read the Introduction (Chapter 2) first, then dip into the lineages in any order.

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Chapter 2

Introduction: The Incestuous Community

> *"I have worked for Thiokol, Morton Thiokol, Thiokol again, ATK Thiokol, ATK Launch Systems, and Northrop Grumman. I have never changed my job."* — Retired engineer, Promontory, Utah, 2019

Learning Objectives

- Understand the scope and methodology of industrial genealogy as applied to the solid rocket sector
- Identify the central thesis: corporate boundaries are porous; technical knowledge and human capital are the durable carriers of institutional memory
- Distinguish between corporate ancestry (formal ownership chains) and practical ancestry (propellant chemistry, facility continuity, personnel networks)

2.1 The Puzzle

By 2020, American strategic ballistic missiles — the Minuteman III ICBMs in their hardened silos across North Dakota, Wyoming, and Montana; the Trident II D5s on Ohio-class submarines beneath the world's oceans — depended for their solid rocket motors on exactly two companies: Northrop Grumman Innovation Systems and Aerojet Rocketdyne.

In 1980, there had been seven. In 1965, there had been more.

The contraction is easy to explain in broad strokes: the Cold War ended, the defense budget fell, and the market could no longer support seven manufacturers of a product the government bought in small quantities at unpredictable intervals. What is harder to explain — and more interesting — is how the knowledge, the facilities, and the people that had been distributed across seven companies ended up concentrated in two.

The answer requires tracing genealogies.

2.2 Corporate Ancestry vs. Practical Ancestry

The distinction that runs through every chapter of this book is between two kinds of ancestry.

Corporate ancestry is the formal chain of ownership and legal succession. Thiokol Chemical Corporation was acquired by Morton International in 1982 to form Morton Thiokol, Inc. Morton Thiokol was spun off as Thiokol Corporation in 1989, then acquired by ATK in 2001. ATK merged with Orbital Sciences in 2015 to form Orbital ATK, which was acquired by Northrop Grumman in 2018. Corporate ancestry can be read off a merger agreement or a SEC filing.

Practical ancestry is the chain through which technical capability actually passes. The propellant formulation for the Minuteman III Stage 1 motor was developed by Thiokol engineers at Promontory, Utah in the early 1960s. Many of those engineers were still working at the same facility — under whatever corporate name appeared on their badge — into the 1990s. The formulation itself, refined but recognizable, was still in production. The facility, expanded and upgraded but continuous, still operated. The corporate name had changed three times. The practical ancestry had not.

This is what "incestuous community" means. Not that the corporate ownership is incestuous — that phrase implies something sinister that the history does not quite support — but that the community of practice is so small, so technically specialized, and so geographically concentrated that it functions, across corporate boundaries, like a single extended family.

2.3 The Seven Lineages

At the Cold War peak, American solid rocket capability resided in seven major corporate entities, each with a distinct genealogical origin:

1. **Thiokol** — chemical origins in polysulfide rubber polymers; Minuteman I and II Stage 1; Space Shuttle SRBs
2. **Hercules** — antitrust breakup of DuPont; Polaris and Trident; Pegasus
3. **Aerojet** — Caltech GALCIT origins; Titan; Minuteman II Stage 2
4. **Atlantic Research Corporation** — independent research origins; JATO and tactical systems
5. **United Technologies Chemical Systems Division** — industrial diversification into aerospace
6. **Olin Aerospace** (née Rocket Research Corporation) — monopropellant origins, tactical systems
7. **Rocketdyne** — primarily liquid rocket heritage, marginal solid capability

Chapters 3 through 5 trace the three major lineages in depth: Thiokol, Hercules, and Aerojet. Chapters 6 through 8 address the "strange bedfellows" — the companies whose primary businesses had nothing obvious to do with rocket propulsion but who owned significant solid rocket assets for years or decades. Chapters 9 through 11 trace the contraction and its consequences.

2.4 Why the Incest Matters

The incestuous character of the community is not merely a colorful historical detail. It has practical implications for understanding the current industrial base — and its vulnerabilities.

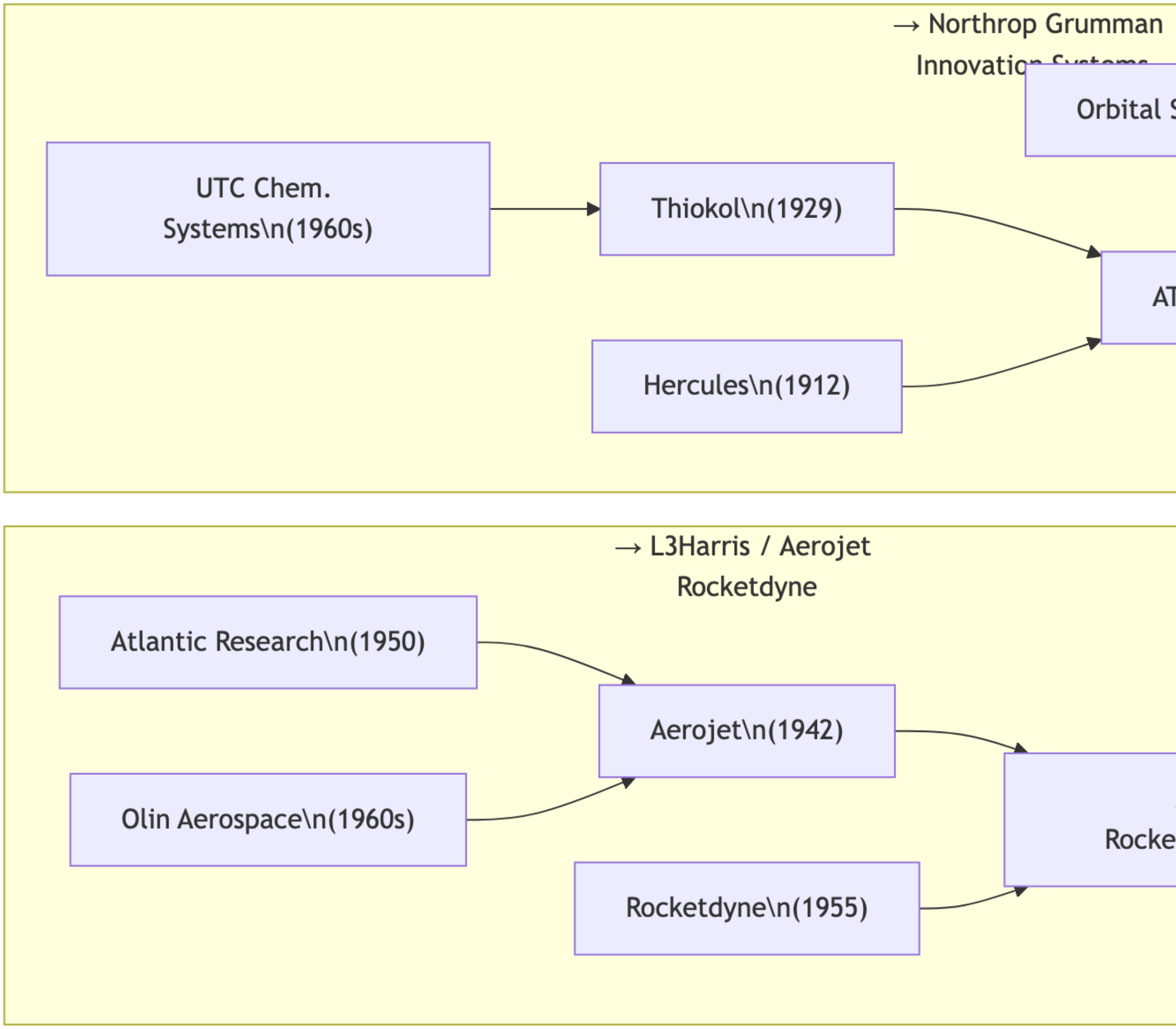

Figure 2.1: Seven Cold War solid rocket motor lineages converging to the present duopoly. UTC CSD was absorbed by Thiokol/Cordant (1996). Olin Aerospace passed through Primex and General Dynamics before reaching Aerojet (2002). ARC was acquired by Aerojet (2003). Rocketdyne merged into Aerojet Rocketdyne (2013).

When Northrop Grumman acquired Orbital ATK in 2018, it acquired not just a legal entity and a set of facilities. It acquired the institutional memory embedded in the workforce, the accumulated process knowledge stored in manufacturing procedures developed over decades, and the technical relationships with government program offices built by individual engineers over careers.

What it could not acquire — or guarantee to retain — was the people who carried that knowledge. The wave of retirements that accelerated as the baby boomer generation aged out of the workforce in the 2010s and 2020s represented a transfer of institutional memory that had no parallel in the merger and acquisition history of the industry.

Corporate genealogy is the record of what was bought and sold. The more important genealogy — harder to trace, impossible to price — is the one that lives in the memory of engineers who have never changed their jobs, only their employers.

2.5 A Note on Sources

This book draws on corporate histories, congressional testimony, trade publications (*Aviation Week & Space Technology*, *Aerospace America*, *Space News*), oral histories conducted by the American Institute of Aeronautics and Astronautics (AIAA) and the Smithsonian National Air and Space Museum, declassified government program records, and secondary historical literature on the defense industrial base.

Where corporate history documents are cited, readers should note that corporate histories are commissioned works and may reflect the perspective of their sponsors. This is particularly true for the merger-era corporate histories produced in the 1990s, which tend to celebrate consolidation as visionary strategy rather than as forced response to demand collapse.

2.6 Summary

The American solid rocket industry is small, technically specialized, and geographically concentrated. Its corporate genealogy — the chains of acquisition, merger, and divestiture that ran from 1950 to the present — is less important than its practical genealogy: the continuous community of engineers, chemists, and program managers who carried its technical knowledge across every corporate transformation. The chapters that follow trace both genealogies, in the hope that the intersection illuminates both.

2.7 Further Reading

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- Spinardi, Graham. *From Polaris to Trident: The Development of US Fleet Ballistic Missile Technology.* Cambridge University Press, 1994.
- Asker, James R. "The Solid Rocket Duopoly." *Aviation Week & Space Technology*, March 2019.
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Chapter 3

The Founding Era: Explosives, JATO, and the Birth of an Industry

> *"What we were trying to do was make a rocket engine that would get a heavily loaded airplane off a short runway. We had no idea what we were starting."* — Frank Malina, GALCIT, 1939

Learning Objectives

- Trace the technical and institutional origins of American solid rocket propulsion from the interwar period through the early Cold War
- Understand why WWII JATO work created the personnel networks that later populated the commercial solid rocket industry
- Identify the government programs — Polaris, Minuteman — that converted research capability into production-scale industrial capacity

3.1 The Explosives Heritage

The American solid rocket industry did not begin with rockets. It began with explosives.

The chemical knowledge, the manufacturing infrastructure, the workforce skills, and the corporate entities that would eventually produce ICBMs and space launch vehicle motors in the 1960s were assembled over the preceding century in the explosives business. Black powder, nitrocellulose, dynamite, smokeless powder — each represented a stage in the accumulation of energetic materials chemistry and processing capability that solid rocket propellants would eventually draw on.

Three facts about the explosives industry shaped its eventual contribution to rocketry:

Production scale. By 1940, American explosives manufacturers had built large-scale energetic materials processing facilities that could be converted — with varying degrees of difficulty — to propellant production. The infrastructure existed. Skilled workers who understood hazardous materials handling, quality control in energetic chemistry, and the organizational discipline that large-scale production of dangerous materials required — these people existed.

Government relationships. Explosives manufacturers operated under close government oversight, particularly after the munitions mobilizations of World War I. These relationships — with the Army Ordnance Department, the Naval Powder Factory, the Bureau of Mines — established the contracting and oversight patterns that defense solid propulsion would inherit.

Geographic concentration. Explosives production clustered in specific regions: New Jersey (DuPont's Carney's Point; Hercules' Kenvil), Maryland (the Naval Powder Factory at Indian Head), Utah (Hercules' Bacchus Works), and California (several facilities). This geographic concentration persists in the solid rocket industry today, for reasons that begin with explosives plant siting logic: remote enough for safety, close enough to rail and labor markets.

3.2 JATO and the GALCIT Circle

The technical origins of American solid rocket propulsion are conventionally traced to the California Institute of Technology's Guggenheim Aeronautical Laboratory (GALCIT) in the late 1930s. The GALCIT Rocket Research Group — eventually formalized as the Air Corps Jet Propulsion Research Project in 1939 — was commissioned to develop Jet-Assisted Take-Off (JATO) units for heavily loaded military aircraft.

The group included figures who would define the first generation of American rocketry: Frank Malina, John Parsons, Edward Forman, Martin Summerfield, and, in an advisory capacity, Theodore von Kármán. Their approach to solid propellant was pragmatic: they needed something that would burn reliably, at predictable thrust, without deflagration-to-detonation transition, and that could be manufactured in the quantities the Army Air Corps wanted.

The solution they arrived at — an asphalt-potassium perchlorate composite propellant, later refined to use ammonium picrate — was crude by later standards but it worked. More important than the chemistry was the community. The GALCIT group was small: perhaps twenty core personnel at its peak. When it dissolved into various successor organizations after the war, those twenty people seeded virtually every significant solid rocket development program in the United States for the next decade.

The most consequential institutional successor was the Jet Propulsion Laboratory, formally established in 1944. But equally important were the private companies that GALCIT alumni founded or joined: Aerojet Engineering Corporation (1942), and the consulting and contracting networks that carried GALCIT methodology into government laboratories and eventually into the commercial sector.

3.3 The WWII Mobilization

The military requirements of World War II transformed what had been a small university research effort into a production industry.

JATO units were needed in large numbers — for carrier aircraft takeoff, for heavy bomber assist, for transport operations in high-altitude or short-field conditions. The Army and Navy contracted with multiple suppliers, including established explosives manufacturers who could ramp production quickly.

The critical technical development of the WWII period was the transition from double-base propellants (nitrocellulose/nitroglycerin formulations borrowed from the smokeless powder industry) to the composite propellants that would eventually enable the large motors the ICBM programs required.

Double-base propellants had been the workhorse of military rocketry since World War I: the German *Nebelwerfer* rockets, the British unguided anti-submarine rockets, the American bazooka. They burned predictably and could be manufactured in the existing smokeless powder infrastructure. But their energy density was limited, and they could not be scaled to the large grain sizes that the postwar strategic programs demanded without severe manufacturing and structural problems.

Composite propellants — a fuel binder combined with an oxidizer in particulate form — offered a path around both limitations. By the late 1940s, ammonium perchlorate (AP) had emerged as the preferred oxidizer, and polyurethane then polybutadiene binders as the preferred fuel-binder systems. The Jet Propulsion Laboratory played a central role in this development; so did Thiokol Chemical Corporation, whose polysulfide rubber binders offered an early alternative that proved well-suited to large-scale production.

3.4 Naval Research and the Polaris Catalyst

The strategic catalyst for industrial-scale solid rocket production was not the Air Force but the Navy.

In the mid-1950s, the Navy faced a fundamental strategic problem: its submarines, the most survivable leg of the nuclear triad, could not carry the ballistic missiles that the Eisenhower administration wanted to field against Soviet cities. The liquid-propellant missiles under development — based on the technology being developed for the Air Force's Atlas and Titan ICBMs — were too large, too volatile, and too hazardous for submarine operation.

The Navy's solution was to gamble on solid propulsion. In 1956, the Polaris Special Projects Office was established under Rear Admiral William Raborn with the explicit mission of developing a submarine-launched ballistic missile using solid propellant motors. Polaris, if it worked, would require solid rocket motors that were radically larger than anything previously manufactured — and that had to meet reliability, safety, and performance requirements more demanding than any previous solid rocket application.

The Polaris program defined the industrial scale of American solid rocket production. It required large-scale manufacturing facilities, quality systems unprecedented in the industry, and a workforce trained in propellant chemistry and processing at levels the existing industry could not supply. The government funded the construction of new facilities — including major expansions at Hercules' Bacchus Works and Aerojet's Sacramento facility — and the training programs that created the workforce.

More important than any single facility was the effect Polaris had on the community of practice. The program pulled engineers and chemists from the explosives industry, from university research groups, and from the small existing JATO manufacturers and concentrated them in a common technical project. The relationships, the informal knowledge networks, and the shared technical vocabulary that resulted were the true founding legacy of Polaris — more durable than any of the hardware.

3.5 Minuteman and the Industrialization of ICBMs

Where Polaris defined the technical scale of solid propulsion, Minuteman — the Air Force's land-based ICBM program initiated in 1958 — defined its industrial scale.

The Minuteman program required thousands of missiles, not hundreds. Its three-stage configuration used solid motors from different manufacturers: Stage 1 from Thiokol, Stage 2 from Aerojet, Stage 3 from Hercules. This division was not accidental. The Air Force, advised by technical experts who had come largely from the GALCIT circle and its successor institutions, structured the program to maintain competitive capability across multiple suppliers — partly to ensure supply security, partly to preserve the competitive pressure that would maintain technical advancement.

The result was that three major companies simultaneously invested in large-scale solid propellant production capability, in parallel with the development of the technical skills and organizational systems that large-scale production required. By the time Minuteman reached full production in the mid-1960s, the American solid rocket industry had the facilities, the workforce, and the institutional capacity to support the strategic arsenal.

The community that Minuteman created — the community that called itself incestuous — was precisely this: the several thousand engineers, chemists, quality specialists, and program managers distributed across Thiokol's Promontory facility, Hercules' Bacchus Works, Aerojet's Sacramento plant, and a dozen ancillary facilities, all connected by the shared technical problems of solid propulsion and by the personnel networks that the Polaris and Minuteman programs had assembled.

3.6 Summary

Period	Key Development	Industrial Consequence
Pre-1940	Explosives industry accumulation	Facilities, skilled labor, government relationships
1939–1945	GALCIT JATO program; WWII mobilization	Technical foundations; personnel network established
1944	JPL formalization	Institutional R&D anchor for the field
1942	Aerojet founding	First dedicated solid/liquid rocket commercial firm
1956–1960	Polaris program	Industrial scale investment; workforce training
1958–1965	Minuteman production	Three-supplier industrial base at production scale

The chapters that follow trace what happened to this industrial base after the Cold War demand that created it began to collapse.

3.7 Further Reading

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- Malina, Frank J. "The GALCIT Rocket Research Project, 1936–1938." *Essays on the History of Rocketry and Astronautics.* NASA, 1977.
- Spinardi, Graham. *From Polaris to Trident.* Cambridge University Press, 1994.
- Neufeld, Michael J. *The Rocket and the Reich.* Harvard University Press, 1995. (Chapter on American technical intelligence and JATO.)
- MacKenzie, Donald. *Inventing Accuracy: A Historical Sociology of Nuclear Missile Guidance.* MIT Press, 1990.

Chapter 4

The Thiokol Lineage: From Rubber Chemistry to the Shuttle and Beyond

"We at Thiokol take our responsibilities very seriously. We are a national resource."
— Edward Stone, Thiokol President, 1986 congressional testimony, three months after Challenger

Learning Objectives

- Trace Thiokol's origins in polysulfide rubber chemistry and its transition to solid propellant production
- Understand the Promontory, Utah facility as the physical locus of the Thiokol lineage across six corporate names
- Analyze the Challenger disaster as a pivotal event in the corporate and regulatory history of the lineage
- Follow the acquisition chain: Thiokol → Morton Thiokol → Thiokol → ATK → Orbital ATK → Northrop Grumman

4.1 Origins: The Rubber Chemist's Gambit (1929)

Thiokol Chemical Corporation was founded in 1929 by Joseph C. Patrick and Nathan Mnookin in Kansas City, Missouri — not in Utah, not in aerospace, and emphatically not in rocketry.

Patrick was a chemist who had been trying to find a synthetic rubber substitute to replace the natural rubber that the automotive industry required in enormous quantities. In experimenting with ethylene dichloride and sodium polysulfide, he produced, by accident, a material with the consistency of rubber but far superior chemical resistance. The material smelled terrible — a sulfur compound — but it held up to solvents, fuels, and industrial chemicals that destroyed natural and synthetic rubbers.

Patrick named it Thiokol (from the Greek *theion*, sulfur, and *kolla*, glue). The company existed to manufacture and sell this polysulfide rubber.

For two decades, it did exactly that. Thiokol polysulfide sealants found markets in aircraft fuel tanks, printing rolls, putties, and coatings — anywhere that solvent resistance and flexibility mattered. The business was solid if unspectacular.

4.2 The Binder That Changed Everything (1947–1955)

The accident of Thiokol's entry into solid propulsion is one of the industry's better-documented stories.

In 1947, Thiokol Chemical was approached by the Army Ordnance Corps, which had been evaluating polysulfide rubber as a potential fuel binder for composite solid propellants. The concept was straightforward: if you could embed an oxidizer (ammonium perchlorate) in a polymer matrix that itself served as fuel, you would have a solid propellant that could be cast rather than pressed — enabling larger grain sizes, better structural integrity, and more uniform burn characteristics than double-base propellants allowed.

Thiokol's polysulfide polymer turned out to be an excellent binder candidate. It was processable at room temperature, it cured reliably, it bonded well to motor cases, and its combustion products were acceptable. By the early 1950s, Thiokol had established a small solid rocket development operation in Elkton, Maryland (at the government-owned Allegany Ballistics Laboratory site) and begun competing for military propulsion contracts.

The first significant program was the Sergeant missile, a tactical surface-to-surface missile for the Army developed in the late 1950s. Sergeant validated Thiokol's production capability and established the Elkton facility as a serious solid propulsion manufacturing site.

4.3 Promontory: The Physical Heart (1956–present)

The Minuteman program required a motor that dwarfed anything Thiokol or anyone else had produced: the Stage 1 motor needed to develop approximately 200,000 pounds of thrust at sea level and burn for approximately 60 seconds. Casting a propellant grain of this size, with reliable burn characteristics, structural integrity under the thermal and mechanical stresses of missile operation, and acceptable case mass fraction, was a manufacturing problem without precedent.

To address it, Thiokol built the Promontory facility.

The site was selected in 1956: a dry lakebed at the north end of the Great Salt Lake desert in Box Elder County, Utah, sufficiently remote for safety, with acceptable seismic stability, rail access, and proximity to a large unskilled and semi-skilled labor market in the Salt Lake City basin. The first permanent structures were completed in 1957.

Between 1957 and 1962, the Promontory facility grew from a greenfield site to the largest solid rocket manufacturing complex in the world. The Minuteman Stage 1 motor — designated the M55 — entered production at Promontory in 1961. The facility employed, at peak Cold War production, more than 10,000 people.

What makes Promontory remarkable is not its Cold War peak but its continuity. The facility that opened in 1957 to build Minuteman motors was still operating in 2024 — under its sixth corporate name — producing solid rocket boosters for the SLS. The same buildings (upgraded and expanded), many of the same production processes (refined but recognizable), and in the early decades, many of the same people.

4.4 The Minuteman Legacy

Thiokol's Minuteman Stage 1 motor program defined the company's technical identity for three decades.

The Stage 1 motor is the largest and most thrust-critical of the three Minuteman stages. Its performance determines the missile's throw weight and range. Thiokol's design — a four-nozzle configuration with a star-perforation grain — became the production baseline that persisted, with periodic performance upgrades, for the full production run of the Minuteman III (which remains operational as of this writing).

The propellant chemistry and grain design knowledge accumulated in the Minuteman program was Thiokol's primary competitive asset through the 1960s and into the 1970s. When NASA began studying large solid rocket boosters for the Space Shuttle in the early 1970s, Thiokol was the technical leader in large-grain composite propellant production — and it won the Shuttle SRB contract in 1973 on that basis.

4.5 The Shuttle SRBs and Elkton

The Space Shuttle Solid Rocket Booster program was the largest single solid rocket motor production contract in history at the time of award.

Each Shuttle flight required two SRBs, each 149 feet long and 12 feet in diameter, generating 3.3 million pounds of thrust at liftoff. Thiokol designed and manufactured the motors at Promontory. The propellant — an HTPB (hydroxyl-terminated polybutadiene) formulation replacing the earlier polysulfide chemistry — was cast in segments, shipped by rail to Kennedy Space Center, and assembled on the launch pad.

The segmented design — four propellant segments joined by field joints sealed with O-rings — was a manufacturing convenience that became, on January 28, 1986, a national catastrophe.

4.6 Challenger and Its Aftermath (1986)

The Challenger accident has been extensively documented — in the Rogers Commission report, in Richard Feynman's famous appendix on O-ring resilience, and in numerous subsequent analyses. The essential finding: the primary O-ring in the right SRB's aft field joint failed to seal at the cold temperatures of launch morning, allowing hot combustion gas to breach the joint and ultimately destroy the external tank.

Less thoroughly documented in the public record — though well known within the solid rocket community — is the character of the management failure that preceded the launch decision. Thiokol engineers had identified O-ring erosion as a concern as early as 1977. The severity of the concern was known at Thiokol; it was reported to NASA; the decision to launch on a January morning when the temperature had fallen to 18°F the night before proceeded over the explicit objections of Thiokol's own engineering team.

The technical failure was O-ring cold resilience. The institutional failure was the management pressure that overrode engineering judgment. Both failures are legible in the structure of the Thiokol-NASA relationship: a sole-source contractor dependent on a single customer, operating under schedule pressure from a program that had promised more flights per year than its vehicle design could safely support.

Thiokol survived Challenger — redesigned the joint, re-qualified the motor, returned to flight in 1988. But the event left a corporate psychology. The name Morton Thiokol, Inc. — the company had been acquired by Morton International in 1982 — became so publicly associated with the accident that Morton spun the aerospace division off as a separate public company in 1989, reverting to the name **Thiokol Corporation**, partly to create a clean separation from the parent's consumer products identity.

4.7 The Corporate Carousel (1929–2024)

The full corporate name sequence of the Promontory facility:

Years	Corporate Name	Owner
1957–1982	Thiokol Chemical Corporation	Independent
1982–1989	Morton Thiokol, Inc.	Morton International
1989–2001	Thiokol Corporation	Independent
2001–2015	ATK Thiokol / ATK Launch Systems	Alliant Techsystems
2015–2018	Orbital ATK	Alliant Techsystems / Orbital Sciences
2018–present	Northrop Grumman Innovation Systems	Northrop Grumman Corp.

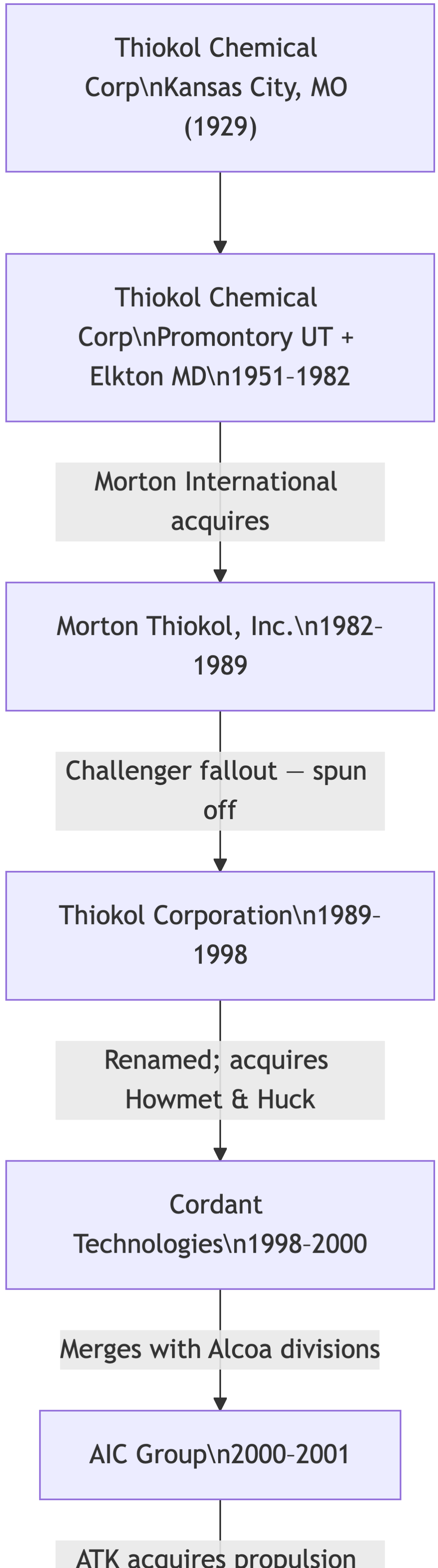
Six corporate names. One facility. The same Wasatch Mountains in the background of every employee badge photo.

4.8 The ATK Acquisition (2001)

Alliant Techsystems (ATK) acquired Thiokol Corporation in 2001 for approximately $685 million. The acquisition is discussed in more detail in Chapter 10; the key point here is what ATK acquired.

Thiokol brought to ATK the following: - The Promontory facility and its production infrastructure for large solid motors - The Elkton, Maryland facility and its tactical propulsion capability - The Minuteman III Stage 1 sustainment contract (ongoing through current day) - The SLS booster development program (then in early stages under NASA's post-Shuttle planning) - The institutional knowledge of large AP-HTPB propellant chemistry and large-grain casting - Approximately 3,000 engineers and skilled production workers

What no acquisition agreement could transfer was the 40-year accumulation of process knowledge held by engineers who had been casting propellant at Promontory since the Minuteman era. This knowledge was the real asset — and its susceptibility to attrition through retirement was a risk that no due diligence document fully captured.


Thiokol Chemical Corp\nKansas City, MO (1929)
Thiokol Chemical Corp\nPromontory UT + Elkton MD\n1951–1982
Morton International acquires
Morton Thiokol, Inc.\n1982–1989
Challenger fallout — spun off
Thiokol Corporation\n1989–1998
Renamed; acquires Howmet & Huck
Cordant Technologies\n1998–2000
Merges with Alcoa divisions
AIC Group\n2000–2001
ATK acquires propulsion

4.9 The SLS Booster: Continuity into the 2020s

The Space Launch System (SLS) five-segment solid rocket boosters represent the direct continuation of the Thiokol technical lineage.

Each SLS booster uses five propellant segments — one more than the Shuttle four-segment design — producing approximately 3.6 million pounds of thrust per booster. The propellant chemistry, the segmented design, the rail shipment from Promontory, and the field assembly at Kennedy Space Center are all direct descendants of the Shuttle SRB program. The manufacturing workforce at Promontory, while substantially renewed through retirements and hiring since 1986, carries institutional knowledge through formal documentation and informal mentorship that traces to the Minuteman production era.

The Artemis program's SLS represents the Thiokol lineage's deepest institutional continuity: a Promontory facility that has been producing solid rocket motors continuously for sixty-seven years, under a corporate name that has changed six times, building hardware whose design ancestry traces directly to Minuteman.

4.10 Summary

The Thiokol lineage illustrates the central argument of this book more clearly than any other. The corporate ownership was turbulent: independent company, division of a salt conglomerate, independent again, defense contractor, merged entity, defense giant. The facility was continuous. The propellant chemistry was continuous. And the people — through every corporate transformation — carried the practical knowledge that the weapons required.

Period	Entity	Key Programs
1929–1956	Thiokol Chemical	Polysulfide rubber, sealants
1957–1965	Thiokol Chemical	Minuteman I/II Stage 1, Sergeant
1965–1982	Thiokol Chemical	Minuteman III Stage 1, early SRB studies
1982–1989	Morton Thiokol	Shuttle SRBs (incl. Challenger)
1989–2001	Thiokol Corp	Shuttle SRBs return to flight, Peacekeeper Stage 1
2001–2018	ATK / Orbital ATK	Shuttle SRBs, SLS development
2018–present	Northrop Grumman	SLS boosters, Minuteman III sustainment

4.11 Further Reading

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Chapter 5

The Hercules Lineage: From Dynamite to Trident

> *"The mining industry in Utah would not have developed if it hadn't been for explosives. If it hadn't been for Goddard, the rocket motor, which again is packaged energy, the whole space thing would not have happened. So the whole history of our country, even the whole history of the world, really evolves around explosives technology."* — G.R. Muir, Hercules Group President, 1988

Of all the lineages in the American solid rocket industry, Hercules is the most improbable. It begins not with a visionary scientist or a government program, but with a federal antitrust lawsuit against a gunpowder monopoly.

5.1 The DuPont Breakup (1912)

E.I. du Pont de Nemours & Company had dominated the American explosives market for over a century. By the early 1900s, DuPont controlled roughly 70% of US explosive powder production. In 1912, a federal court ordered DuPont to divest, creating two entirely new companies: Atlas Powder Company and Hercules Powder Company.

Hercules received DuPont's black powder and dynamite businesses. Almost immediately, it built the **Bacchus Works** in 1913 in Magna, Utah — chosen for its proximity to the region's copper and silver mines. No one imagined that the Bacchus Works would still be operating a century later, producing propellant for nuclear-armed submarine missiles.

5.2 The Bacchus Works: One Site, Six Corporate Owners, 110 Years

The Bacchus Works is the physical spine of the Hercules story:

1913 — Hercules Powder Company opens the site to produce dynamite for Utah's mining industry.

1940 — Kenvil, NJ disaster. On September 12, the flagship New Jersey plant suffered a catastrophic explosion — 15 buildings destroyed, 51 killed, hundreds injured.

1944 — Hercules began operating the Allegheny Ballistics Laboratory at the Naval Powder Factory near Cumberland, Maryland. Some 700 scientists, engineers, and technicians worked on solid propellant rocket development — the foundation for everything that followed.

1959 — Between 1955 and 1963, Hercules saw its sales double, due in large part to government contracts. In 1959, Hercules diversified into rocket fuels and propulsion systems for the Polaris, Minuteman, and Honest John missiles. Sales of aerospace equipment and fuels accounted for almost 10% of sales in 1961, 15% in 1962, and 25% in 1963.

1961 — "Air Force Plant 81" expansion: 97 new buildings added to Bacchus Works, including a 97,000 square foot administration building.

1966 — Renamed **Hercules, Inc.** as the company's identity shifted decisively from explosives to aerospace.

1978 — Dedicated **Aerospace Division** created to house growing missile and space propulsion work.

1980 — Hercules bought filament-winding technology from a plant in England and adapted it for graphite fibers — stronger than steel, lighter than titanium. Engineers increased the fibers' ability to withstand pressure from 200 pounds per square inch to 1 million psi. This innovation dramatically reduced motor weight and became Hercules's signature technical advantage.

1989 — Hercules co-funded the **Pegasus** air-launched rocket with Orbital Sciences Corporation, building all three solid rocket stages — the world's first privately developed orbital launch vehicle stages.

1995 — ATK purchased the Hercules Aerospace Division. The Hercules Composite Product Division was purchased separately by Hexcel Corporation in 1996.

2015/2018 — ATK merged with Orbital Sciences to form Orbital ATK, then Northrop Grumman acquired Orbital ATK for $7.8B. The Bacchus Works now operates under Northrop Grumman Innovation Systems.

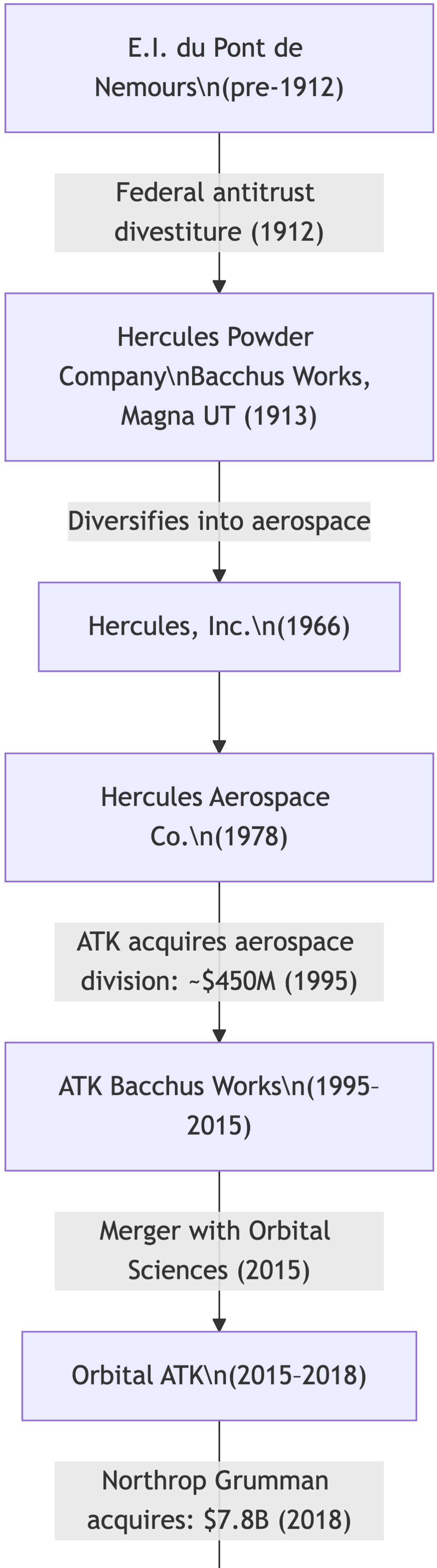
5.3 The Program Portfolio

Hercules's program history reads like a complete catalog of American strategic deterrence:

Polaris (Navy, 1950s–1970s): Hercules produced rocket motors for the US Navy's 41 for Freedom series — 41 George Washington-class ballistic missile submarines, each carrying 16 Polaris missiles, 656 missiles total. The Polaris gave the United States an invulnerable second-strike capability that fundamentally changed Cold War nuclear strategy.

Minuteman III — Stage 3 (Air Force, 1960s–present): The third stage of the Minuteman III was Hercules's domain throughout the ICBM program's production run. This heritage now lives in Northrop Grumman's portfolio.

Poseidon and Trident II D5 (Navy, 1970s–present): Northrop Grumman now manufactures solid propulsion boost motor systems for all three stages of the Trident II D5 — the current backbone of the US Navy's nuclear deterrent. That capability traces directly to Hercules's Bacchus Works work from the 1970s onward.


E.I. du Pont de Nemours\n(pre-1912)
Federal antitrust divestiture (1912)
Hercules Powder Company\nBacchus Works, Magna UT (1913)
Diversifies into aerospace
Hercules, Inc.\n(1966)
Hercules Aerospace Co.\n(1978)
ATK acquires aerospace division: ~$450M (1995)
ATK Bacchus Works\n(1995-2015)
Merger with Orbital Sciences (2015)
Orbital ATK\n(2015-2018)
Northrop Grumman acquires: $7.8B (2018)

Titan IV Strap-On Boosters: Hercules developed the large strap-on solid boosters for the Titan IVB — the largest unmanned rocket available after Saturn V was retired — which carried Cassini to Saturn and classified NRO payloads to orbit.

Pegasus (1989–present): The three Orion solid motors built for Orbital Sciences were the first privately funded orbital launch vehicle stages in history. Pegasus flew its final mission in 2021, with one more planned for 2026.

5.4 The Nobel Laureate Connection

One footnote deserves mention: Richard F. Heck, recipient of the 2010 Nobel Prize in Chemistry, gained experience with transition metal chemistry while working at Hercules in 1957. Heck won the Nobel for palladium-catalyzed cross coupling reactions — now a foundational tool in pharmaceutical synthesis. The same company making Polaris missile motors was employing the chemist who would later revolutionize drug manufacturing.

5.5 The Incest Angle

Consider the career arc available to a Hercules engineer who joined in 1960:

- Polaris propellant work at Kenvil or Bacchus
- Minuteman Stage 3 production at Bacchus Works

- Graphite composite casing development in the 1980s
- Pegasus motor design, 1987–1989
- Transfer to ATK Bacchus when the aerospace division was sold in 1995
- Retirement from Orbital ATK or early Northrop Grumman

Their employer's name changed five times. Their facility, their colleagues, their propellant chemistry — largely continuous. This is the incest: the people persist long after the corporate names change.

5.6 Summary

Period	Entity	Key Programs
1913–1959	Hercules Powder Co.	Dynamite, WWII munitions
1959–1966	Hercules Powder Co.	Polaris, Minuteman III Stage 3, Honest John
1966–1978	Hercules, Inc.	Polaris, Poseidon, Minuteman, Pershing
1978–1995	Hercules Aerospace Co.	Trident II, MX/Peacekeeper, Titan IV, Pegasus
1995–2015	ATK Bacchus Works	Trident II, SLS development, Pegasus
2015–2018	Orbital ATK	Trident II, SLS, Antares, Pegasus
2018–present	Northrop Grumman Innovation Systems	Trident II D5, Sentinel, SLS

5.7 Further Reading

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- Hunley, J.D. *The Development of Propulsion Technology for U.S. Space-Launch Vehicles, 1926–1991*. Texas A&M University Press, 2007.
- Harvey, Brian. *Russia in Space: The Failed Frontier?* Springer/Praxis, 2001. (Useful comparative perspective on how Soviet solid propulsion developed in parallel.)
- United States Department of Justice. *United States v. E.I. du Pont de Nemours & Co.*, 188 U.S. 127 (1911). (The antitrust judgment that created Hercules Powder Company.)

5.8 Exercises

1. The Bacchus Works has operated under six corporate owners since 1913. What factors — physical, institutional, contractual — explain this continuity when so much else has changed?

2. Hercules's pivot from dynamite to rocket propellant occurred over roughly four years (1959–1963). What conditions made this transition possible?

3. The Trident II D5 flies on propulsion with direct lineage to Hercules's 1970s–80s work. What are the implications of running nuclear deterrent systems on propulsion technology developed 40–50 years ago?

4. Richard Heck worked at Hercules in 1957 during its early rocket work. Research another instance where defense-industrial research produced significant civilian scientific spillovers.

Chapter 6

The Aerojet Lineage: From JATO to Rocketdyne

> *"Aerojet was called the 'General Motors of U.S. Rocketry' by Time magazine in 1958; five years later, the company employed 34,000 people working on missiles such as the Polaris, Minuteman, Trident, and Titan. Revenues were $605 million in 1962."* — Encyclopedia.com

The Aerojet lineage is the most scientifically distinguished of the major solid rocket families, born directly from academic rocket research at Caltech. It is also the most administratively tangled — passing through the hands of a tire company, an airline operator, a film studio owner, a chemicals conglomerate, and finally a defense electronics giant, all while producing propulsion systems for America's most critical missiles.

6.1 Origins: Caltech, JATO, and von Kármán (1942)

Aerojet Engineering Corporation was co-founded on March 19, 1942 by a remarkable group of seven — among them Theodore von Kármán, the Hungarian-American aerodynamicist who was perhaps the most important figure in American aerospace in the mid-20th century, along with Frank Malina, Andrew Haley, Martin Summerfield, and Jack Parsons, all connected to Caltech and the Jet Propulsion Laboratory.

The company's first product was the Jet-Assisted Take-Off (JATO) unit — a solid or liquid rocket strapped to an aircraft to help it lift off from short runways or carrier decks. The demo was a success. The company landed its first production contract in 1942 and began its ascent.

Almost immediately, a critical early partnership formed. Aerojet needed a rubber binder for its solid propellant formulations. It turned to **General Tire & Rubber Company** for the chemistry. General Tire became a major Aerojet shareholder — the first of what would become a long series of strange bedfellow relationships. A tire company had bought into the rocket business.

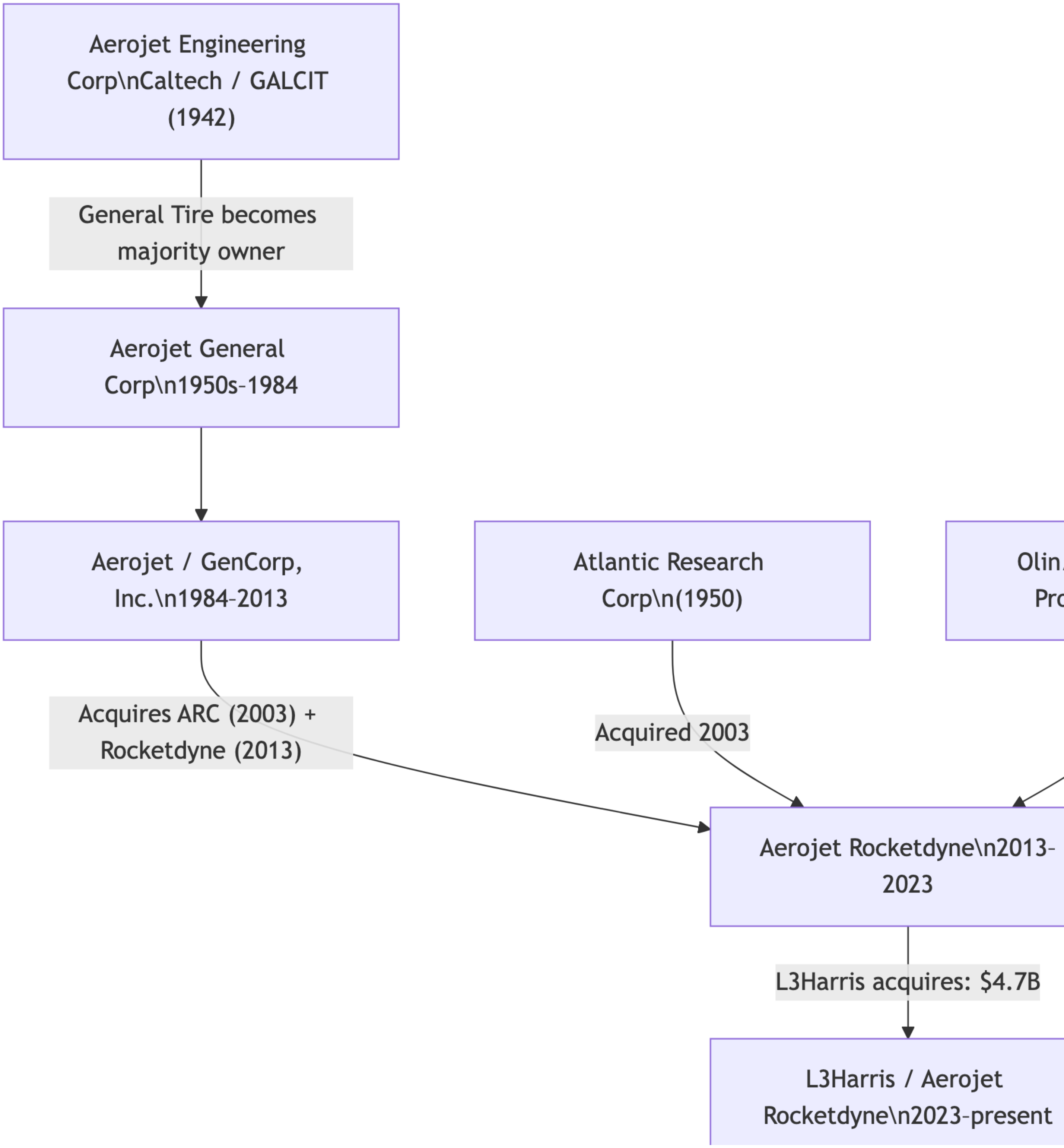

Figure 6.1: Aerojet ownership chain. Atlantic Research (1950) and Olin/GD Space Propulsion joined via acquisition. Rocketdyne merged in 2013.

6.2 Sacramento: The Gold Fields Become a Rocket Range (1950–1953)

In 1950, Aerojet acquired 7,300 acres just east of Sacramento, California. The area had been dredged for gold by the Natomas Mining Company, leaving large earthen berms perfectly suited for rocket test cells — the berms provided natural blast protection. Aerojet converted these former gold fields into one of the premier rocket manufacturing and testing facilities in the world.

In March 1951, Aerojet moved its solid rocket motor production from Azusa to Sacramento. By 1953, the company had established a weapons plant in Rancho Cordova. The Sacramento complex — eventually encompassing approximately 12,600 acres — would remain Aerojet's primary manufacturing and testing location for decades.

6.3 The ICBM Boom: 34,000 Employees and $605 Million (1958–1963)

The US Air Force selected Aerojet as a primary supplier on a number of ICBM projects, including the Titan and Minuteman missiles. They also delivered propulsion systems for the Navy's Polaris submarine-launched ballistic missile.

The growth was staggering. Aerojet employed nearly 2,800 people in 1952, with sales of $21 million. By 1963, the company employed 34,000 people, with revenues of $605 million — a 12-fold increase in headcount and nearly 30-fold in revenue in eleven years. Time magazine called it the "General Motors of U.S. Rocketry" in 1958.

In 1959, the company created two new divisions — Ordnance and Electronic Systems. The Electronic Systems Division created infrared technology allowing satellites to observe missile launches around the world — a vital early warning capability.

6.4 Key Programs

Titan ICBM and Launch Vehicle: The Titan launch vehicle family was established in October 1955 when the US Air Force awarded a contract for a heavy-duty ICBM. Aerojet designed and built a total of 1,182 engines for all four incarnations of the Titan rockets, used for everything from Gemini crewed flights to the Cassini, Viking, and Voyager planetary missions. The Titan launch vehicle was retired from service in mid-1987 — one of Aerojet's longest-running program relationships.

Minuteman — Stage 2: Aerojet held the Stage 2 contract on the Minuteman ICBM, alongside Thiokol (Stage 1) and Hercules (Stage 3). This three-company division of the most important American ICBM program defined the competitive landscape for decades.

Polaris and Poseidon: Aerojet delivered solid propulsion systems for the Navy's submarine-launched ballistic missile programs alongside Hercules, reinforcing its position as a Navy propulsion supplier.

Peacekeeper (MX) — Stage 2: In April 1978, contracts for the four Peacekeeper stages were distributed across the established industrial base — Thiokol, Aerojet, Hercules, and Rocketdyne. Aerojet's second stage SR119 solid fuel motor served from 1986 to 2005 when the Peacekeeper was retired.

Space Shuttle Orbital Maneuvering System (OMS): Aerojet produced the Orbital Maneuvering System and Reaction Control System engines for the Space Shuttle — the liquid propellant engines that maneuvered the Orbiter in space and controlled its attitude. While Thiokol built the solid boosters, Aerojet powered the spacecraft itself.

Advanced Solid Rocket Motor (ASRM): In the early 1990s, Aerojet pursued the contract for Advanced Solid Rocket Motors for the Shuttle — a program eventually cancelled after the Cold War budget contraction.

6.5 The General Tire / GenCorp Parentage: Tires, Films, Airlines, Rockets

The Aerojet lineage is inseparable from its parent company's extraordinary diversification. General Tire & Rubber Company — founded in Akron, Ohio in 1915 — accumulated one of the most eclectic corporate portfolios in American business history while holding Aerojet as a subsidiary:

- **1942:** Invested in Aerojet Engineering Corporation, becoming a major shareholder
- **1942:** Bought Yankee Network, a Boston-based chain of radio stations
- **1945:** Switched part of production to defense with the Aerojet investment
- **1948:** Moved into TV, going on air at WNAC-TV Boston
- **1955:** Purchased RKO Pictures from Howard Hughes — gaining a movie studio, film library, and theatrical chain
- **1958:** Sold the movie business to Desilu (Lucille Ball and Desi Arnaz), retaining radio and TV holdings
- **1964:** Acquired Frontier Airlines
- **1965:** First hotel purchase
- **1984:** Renamed **GenCorp, Inc.**, restructuring as a holding company

At its peak of diversification, GenCorp was simultaneously operating: Aerojet (rockets and missiles), Frontier Airlines (commercial aviation), General Tire (tires), television and radio stations, hotels, and specialty chemicals. The company that supplied solid rocket stages for Minuteman and Titan was also flying passengers between Denver and Albuquerque.

GenCorp eventually unwound this diversification. It sold General Tire to German manufacturer Continental AG to concentrate on Aerojet. It sold Frontier Airlines. It divested its decorative and building products and performance chemicals businesses into OMNOVA Solutions. By the 2000s, GenCorp's two remaining businesses were Aerojet and Easton Real Estate — the latter managing the enormous Sacramento land holdings.

6.6 The Consolidation: Acquiring Atlantic Research and General Dynamics Space Systems

As the Cold War ended and the defense market contracted, Aerojet pursued a consolidation strategy to maintain scale:

2002: Aerojet acquired the Space Propulsion and Fire Suppression business from General Dynamics Ordnance and Tactical Systems — adding spacecraft propulsion capabilities including monopropellant hydrazine thrusters, bipropellant engines, and electric propulsion.

2003: Aerojet acquired **Atlantic Research Corporation** — the Virginia-based company that had pioneered aluminum composite propellant chemistry in 1954. ARC's contribution to solid propellant science was foundational; acquiring it brought those capabilities in-house. Aerojet's rocket engine for the Delta II second stage completed a record 268 successful mission launches since 1960.

6.7 The Rocketdyne Merger (2013): Two Lineages Become One

In July 2012, GenCorp agreed to buy Pratt & Whitney Rocketdyne from United Technologies Corporation for $550 million. The FTC approved the deal in June 2013, and it closed June 17. GenCorp immediately merged Rocketdyne with Aerojet to form **Aerojet Rocketdyne** — combining what had been two separate lineages: the Caltech/JATO/General Tire heritage with the North American Aviation/V-2/liquid engine heritage.

On April 27, 2015, the corporate name was officially changed from GenCorp, Inc. to **Aerojet Rocketdyne Holdings, Inc.**

6.8 The Lockheed Attempt and L3Harris (2020–2023)

On December 20, 2020, it was announced that Lockheed Martin would acquire Aerojet Rocketdyne for $4.4 billion. The FTC sued to block the deal on a 4-0 vote in January 2022, ruling that the acquisition would eliminate the largest independent maker of rocket motors and give Lockheed unfair advantage over competitors who depended on Aerojet for propulsion. Lockheed abandoned the deal in February 2022.

In December 2022, L3Harris Technologies agreed to buy Aerojet Rocketdyne for $4.7 billion. The acquisition completed in July 2023. L3Harris — a defense electronics and communications company formed from the 2019 merger of L3 Technologies and Harris Corporation — became the parent of the second major US solid rocket motor manufacturer, completing the industry's consolidation into a duopoly.

Then, in early 2017, Aerojet had announced it would cease manufacturing in Rancho Cordova, relocating or eliminating about 1,100 of its 1,400 local jobs by 2019. The Sacramento complex — built on gold dredge tailings in 1950, home to 20,000 workers during the 1960s space race — was being wound down. The Sacramento Bee wrote: "The company that arrived here in 1951 and once dedicated more than 20,000 workers to the 1960s space race and the gigantic history of putting men on the moon will soon be a shell of what it once was."

6.9 The Incest Angle: Competing Divisions of the Same Program

The Aerojet lineage's most pointed illustration of industry incest is the Minuteman program itself. Three companies — Thiokol, Aerojet, Hercules — each held one stage of the same missile and competed fiercely for production contracts. Their engineers attended the same conferences, published in the same journals, worked on the same propellant chemistry challenges. Some left one company for another. The technical boundaries were real but porous; the human network was shared.

By 2018, all three Minuteman propulsion lineages had been absorbed into a single corporate entity: Northrop Grumman Innovation Systems. The competition was over. The incest was complete.

6.10 Summary

Period	Entity	Parent	Key Programs
1942–1950s	Aerojet Engineering Corp.	General Tire (shareholder)	JATO, early solid motors
1950s–1984	Aerojet General Corp.	General Tire & Rubber	Titan, Minuteman II, Polaris, Shuttle OMS
1984–2013	Aerojet (GenCorp subsidiary)	GenCorp, Inc.	Peacekeeper Stage 2, Delta II, ARC acquisition
2013–2023	Aerojet Rocketdyne	GenCorp / Aerojet RD Holdings	Merged with Rocketdyne; RS-25, RL-10, solid motors
2023–present	Aerojet Rocketdyne	L3Harris Technologies	GBI, Sentinel competition, RL-10, RS-25

6.11 Further Reading

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- Koppes, Clayton R. *JPL and the American Space Program: A History of the Jet Propulsion Laboratory.* Yale University Press, 1982. (Context for Aerojet's Caltech/GALCIT origins.)
- Federal Trade Commission. *In the Matter of Lockheed Martin Corporation and Aerojet Rocketdyne Holdings, Inc.* FTC Docket No. 9408, 2022. (The FTC complaint blocking the Lockheed acquisition.)

6.12 Exercises

1. Aerojet grew from 2,800 employees and $21M revenue in 1952 to 34,000 employees and $605M revenue in 1963. Plot this growth curve and compare it to the growth rates of other defense companies during the same period. What drove this exceptional rate?

2. General Tire's portfolio at peak diversification (tires, rockets, radio stations, a movie studio, an airline, hotels) seems absurd by today's corporate governance standards. Research the conglomerate era of American business (1960s–1980s) and explain why this kind of diversification was considered rational strategy at the time.

3. The FTC blocked Lockheed's acquisition of Aerojet Rocketdyne on competition grounds — ruling that Lockheed would disadvantage competitors who relied on Aerojet propulsion. Analyze this decision. Was the FTC right? What would the industry look like today if the deal had closed?

4. The Sacramento complex — 12,600 acres built on gold mine tailings, once employing 20,000 people — is now largely being wound down. Research what happens to large defense-industrial sites when programs end. What are the environmental, economic, and community implications?

Chapter 7

Strange Bedfellow #1: Morton Salt and the Shuttle Boosters

> *"The shuttle thing will cost us ten cents a share."* — Morton Thiokol Chairman Charles Locke, to the Wall Street Journal, January 28, 1986

Of all the strange bedfellow stories in the solid rocket industry, none is stranger, more consequential, or more human than Morton Salt's seven-year ownership of the company that built the Space Shuttle's solid rocket boosters.

It begins, as so many American corporate stories do, with a hostile takeover threat.

7.1 Morton in 1982: Needing Armor

The Morton Company — probably best known for its popular blue canister of table salt, featuring a raincoat-clad girl with an umbrella and the tagline "When it rains, it pours" — traced its origins to 1848, when Joy Morton began distributing salt in Chicago from Erie Canal shipments. After the Civil War, the demand for salt expanded with the meat-packing industry, and Morton's Chicago operation grew into a major national business.

By 1982, Morton-Norwich Products (the company had diversified into pharmaceuticals and household products through acquisition) was in financial trouble — not because its businesses were performing poorly, but because of a structural vulnerability. Morton's steady income from salt made its stock attractive and predictable. Its shares were selling for $30, but the company only held $20 per share in cash reserves. In the language of 1980s corporate raiders, Morton was a target.

Management at Morton decided to act. They needed debt on their balance sheet — the counterintuitive defensive strategy of making the company expensive and messy enough to deter acquirers. They also needed growth. In 1982, Morton sold the Norwich pharmaceutical division and used part of the proceeds to buy back stock from Rhone-Poulenc, a French chemical company that had accumulated a stake.

But that wasn't enough. They needed a merger partner.

7.2 Thiokol in 1982: An Attractive Target

Thiokol Chemical Corporation, by 1982, was one of the most successful specialty chemical and defense companies in the United States:

- **40% of the solid rocket fuel market** — a near-dominant position
- **20% annual growth rate** — exceptional for any industrial company
- **Space Shuttle SRB contract** — awarded in 1974, the company was the sole manufacturer of the reusable solid rocket boosters for NASA's flagship program
- **Texize division** — a profitable line of household cleaners including Glass Plus

Morton's management believed the household products divisions of both companies would complement one another. Morton's salt and specialty chemicals worked alongside Thiokol's Texize cleaners. And Thiokol's defense cash flows — reliable, government-backed, growing under the Reagan defense buildup — would stabilize Morton against future takeover threats.

In 1982, Morton completed the acquisition of Thiokol. The merged entity became **Morton Thiokol, Inc.** The company with the little girl and the umbrella now owned the solid rocket boosters for the Space Shuttle.

7.3 The First Year: Management Walks Out

The merger's first year revealed the cultural incompatibility that would shadow the partnership. Morton's management culture — the "bean counters" as Thiokol's engineers called them — clashed immediately with Thiokol's tradition of technological entrepreneurship.

A severe disagreement arose between the upper management of each company. The result: Thiokol's president Robert Davies — considered one of the brightest executives in the aerospace and chemical industries — walked out. Four other high-level executives with experience in aerospace followed, either retiring or resigning. Morton's Charles Locke was given complete control of both companies.

Despite these defections, the company posted record earnings in its first year. Earnings per share increased 26% two years after the merger. Morton's and Thiokol's specialty chemicals divisions worked well together. The financial logic held — the human logic did not.

7.4 January 28, 1986: Challenger

At 11:38 AM Eastern time on January 28, 1986, the Space Shuttle Challenger lifted off from Launch Complex 39B at Kennedy Space Center. Outside air temperature was 36°F — far below the 40°F minimum rating for the solid rocket booster O-rings.

At 73 seconds after launch, a flame from the right solid rocket booster's aft field joint burned through the external tank, causing structural failure and breakup of the vehicle. All seven crew members — Francis Scobee, Michael Smith, Judith Resnik, Ellison Onizuka, Ronald McNair, Gregory Jarvis, and Christa McAuliffe — were killed.

The investigation revealed what Morton Thiokol's own engineers had known and warned about: the O-ring seals were not rated for operation below 40°F, and engineers had specifically recommended against the launch the previous evening. They were overruled.

The Presidential Commission on the Space Shuttle Challenger Accident (the Rogers Commission) found that Morton Thiokol management had reversed its engineers' recommendation under pressure from NASA managers, who were eager to launch. Roger Boisjoly, a Morton Thiokol engineer, had written a prescient memo months earlier warning of exactly this failure mode. His warnings were not acted upon.

7.5 Locke's Response: Ten Cents a Share

Morton Thiokol chairman Charles Locke's response to the disaster became one of the most infamous quotes in American corporate history. Shortly after the accident, he told reporters that "the shuttle thing will cost us ten cents a share." Quoted out of context, the remark nonetheless ensured Locke a reputation as tactless and insensitive that persisted for the rest of his career.

The fuller context was not much better: Locke ordered an investigation of the accident but failed to handle the public relations crisis that followed. The company was on the wrong side of the largest peacetime aerospace disaster in American history, and its chairman was talking about earnings per share.

7.6 The Divorce: 1989

Despite performing $400 million worth of redesign work largely at cost, and despite the resumption of Shuttle flights in 1988, Morton Thiokol lost a bid for a new booster design to Lockheed. At this point, Locke decided to spin off the Thiokol division.

Before dividing the companies, Locke executed a maneuver that Thiokol employees viewed as a betrayal: he transferred Thiokol's most profitable non-rocket businesses to the Morton side of the company. Morton retained:

- **The specialty chemicals business** — adhesives, coatings, sealants, electrical chemicals, dyes
- **The airbag business** — a promising automotive safety technology that would grow to $269M in revenue by 1994

The companies were officially split on July 1, 1989. Morton International, with $1.4 billion in sales and 8,000 employees, remained concentrated in chemicals and salt. Utah-based Thiokol Corporation emerged as a $1.1 billion company with 12,000 employees — but its profitable chemicals and airbag businesses had been carved away. All that remained was rockets.

7.7 The Airbag Aftermath: Salt to Airbags

The airbag business Morton retained from Thiokol grew into a substantial operation. Morton's airbag revenue reached $269M by 1994 — making Morton International, the salt company, briefly one of the world's largest automotive safety suppliers. In the late 1990s, Morton sold the airbag business to Autoliv, the Swedish automotive safety company.

The causality chain is remarkable: Morton Salt acquired Thiokol to protect itself from hostile takeovers. Thiokol's airbag patent (Thiokol had been awarded the first patent for a "safety cushion assembly for automotive vehicles" in 1953) was among the assets Morton retained. That airbag business, grown and sold to a Swedish company, was the most financially successful legacy of the entire Morton-Thiokol merger.

The rocket business ended up costing Morton. The airbag business made money.

7.8 Legacy: What the Marriage Cost, and What It Produced

The Morton Thiokol period (1982–1989) produced:

- The Space Shuttle solid rocket booster redesign after Challenger (though at tremendous cost)
- Seven crew member deaths and the 32-month grounding of the Shuttle program
- The dismissal of Thiokol's senior technical management
- Long-term damage to NASA's safety culture (documented in subsequent investigations including Columbia in 2003)
- The demonstration that defense cash flows cannot easily be managed by non-defense corporate cultures

It also produced, inadvertently: - A survivor company (Thiokol Corporation) that eventually became the propulsion core of ATK - The airbag industry heritage that Morton took with it - One of the most studied corporate ethics cases in American business school curricula

Roger Boisjoly, the Thiokol engineer who warned about the O-rings, spent years afterward speaking publicly about the ethical failure at Morton Thiokol. He framed it not as an engineering failure but as a management failure — the moment when financial and schedule pressure overrode the judgment of the people who actually understood the hardware.

The solid rocket industry has not forgotten it. Every program review, every launch readiness review, every time an engineer raises a concern about flight conditions — Challenger is the institutional memory that explains why those processes exist.

7.9 Further Reading

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- Presidential Commission on the Space Shuttle Challenger Accident. *Report to the President*. 1986. (The Rogers Commission Report.)
- Bazerman, Max H., and Ann E. Tenbrunsel. *Blind Spots: Why We Fail to Do What's Right and What to Do about It*. Princeton University Press, 2011. (Business ethics framework applied in many Challenger case analyses.)

7.10 Exercises

1. Reconstruct the Morton management's decision logic for acquiring Thiokol in 1982. Given what they knew at the time, was it a reasonable strategic decision? What information

would have changed the calculus?

2. The Rogers Commission found that Morton Thiokol management reversed an engineering recommendation under pressure from NASA. Research the concept of "normalization of deviance" as described by sociologist Diane Vaughan in her book on Challenger. How does this concept explain what happened?

3. Morton retained Thiokol's airbag business when it spun off the rocket division in 1989 — and that business proved more valuable than the rockets. What does this suggest about how diversified companies evaluate subsidiary assets during divestitures?

4. Roger Boisjoly warned about the O-rings in writing months before Challenger. His warning was not acted upon. Research what organizational structures or processes have been implemented in the aerospace industry since Challenger specifically to prevent the suppression of engineer safety concerns.

Chapter 8

Strange Bedfellow #2: General Tire — Tires, Films, Airlines, and ICBMs

> *"By 1963, Aerojet employed 34,000 people working on missiles such as the Polaris, Minuteman, Trident, and Titan."*

If Morton Salt's ownership of rocket boosters is the industry's most consequential strange bedfellow story, General Tire & Rubber Company's ownership of Aerojet is the most absurd. At the height of the Cold War, the same parent company was simultaneously making tires, running radio stations, operating a movie studio bought from Howard Hughes, flying a commercial airline, and building stages for America's ICBMs.

The story begins, like so many in this industry, with rubber chemistry.

8.1 The Rubber Connection (1942)

When Aerojet Engineering Corporation was founded in 1942, its co-founders needed a binder for their early solid propellant formulations — a material that could hold oxidizer and fuel together in a stable, castable matrix. They turned to General Tire & Rubber Company of Akron, Ohio, which had the rubber chemistry expertise they needed.

General Tire provided both the technical knowledge and startup capital, becoming a major Aerojet shareholder. For General Tire's management, the investment seemed logical: they were a rubber and materials company; Aerojet needed rubber chemistry expertise; the relationship made technical sense. The defense contracts provided a steady income stream independent of the civilian tire market.

What no one anticipated was that this practical investment would eventually make General Tire the parent company of the dominant US liquid and solid rocket motor manufacturer.

8.2 The Diversification Empire

Through the 1940s and 1950s, General Tire's management pursued a diversification strategy of staggering breadth. The company's SEC filings from this era read like a fever dream of mid-century American capitalism:

Media Empire: - **1942:** Acquired the Yankee Network, a Boston-based chain of radio stations - **1948:** Moved into television, going on air at WNAC-TV Boston - **1955:** Purchased RKO Pictures from Howard Hughes — acquiring not just a film studio, but an entire film library (including King Kong and Citizen Kane), theatrical chains, and an international distribution network - **1958:** Sold the movie business to Desilu Productions (Lucille Ball and Desi Arnaz) — retaining the radio and television holdings

Transportation: - **1964:** Acquired Frontier Airlines — a regional carrier operating primarily in the mountain west

Hospitality: - **1965:** First hotel purchase — beginning what became a small chain

Core Manufacturing: - Tires (the original business, eventually sold to Continental AG) - Synthetic rubber plants

Defense/Aerospace: - Aerojet Engineering Corporation (major shareholder from 1942)

The same boardroom that made decisions about the RKO film library and Frontier Airlines flight schedules was also overseeing the development of propulsion systems for Minuteman and Titan missiles. This was not unusual for the era — the conglomerate model was the dominant corporate strategy of the 1960s — but its application to rocket propulsion creates a particularly vivid illustration.

8.3 Why This Made (Some) Sense

The conglomerate model of the 1960s was not irrational, despite how it looks in hindsight. The logic:

Diversification reduces risk. A company with 10 unrelated businesses is less exposed to any single market downturn than a company with one. The tire business was cyclical. Defense contracts were steady. Media revenues were growing. The combination theoretically created a more stable entity.

Capital allocation. A profitable core business (tires) generates cash that can be deployed into higher-growth businesses (defense, media). The parent becomes a capital allocator rather than a pure operator.

Conglomerate premium. In the 1960s, Wall Street rewarded conglomerates with premium valuations, believing that professional management could run any business efficiently. This premium eventually collapsed in the 1980s as the weaknesses of the model became apparent.

What the model could not adequately handle was the operational reality of managing genuinely different businesses. Running an airline requires different expertise than making tires, which requires different expertise than managing film distribution rights, which requires different expertise than overseeing rocket propulsion development. General Tire's management eventually had to acknowledge this reality.

8.4 The Unwind: GenCorp (1984)

In 1984, General Tire & Rubber renamed itself GenCorp, Inc. and restructured as a holding company — acknowledging that it was a portfolio of businesses rather than a unified operating company. The new structure was meant to allow each subsidiary to operate more independently.

The tire business was eventually sold to Continental AG of Germany — the irony of an American tire company selling to a German competitor was not lost on observers. Frontier Airlines went through deregulation and was eventually absorbed into other carriers. The media properties were sold.

By the late 1990s and 2000s, GenCorp's two remaining businesses were Aerojet and Easton Real Estate — the latter managing the enormous Sacramento land holdings Aerojet had accumulated for its test facilities.

8.5 The Final Chapter: Tires to Rocketdyne

In July 2012, GenCorp agreed to buy Pratt & Whitney Rocketdyne from United Technologies Corporation for $550 million. The acquisition completed in June 2013, and GenCorp immediately merged Rocketdyne with Aerojet to form Aerojet Rocketdyne. The corporate name changed to Aerojet Rocketdyne Holdings in 2015.

A company that started as a tire manufacturer in Akron in 1915 had, over a century, become the parent of the combined Aerojet and Rocketdyne propulsion heritage — two of the most significant lineages in American rocket history. The tire company owned the engines that went to the moon.

L3Harris acquired Aerojet Rocketdyne in 2023 for $4.7 billion, ending the General Tire lineage's independence.

8.6 The Incest Angle: Competing Alongside Your Parent

The General Tire / Aerojet relationship illustrates another dimension of the industry's incestuous character: the parent-child competitive dynamic. Aerojet competed for solid rocket motor contracts against Thiokol and Hercules throughout the 1960s–1990s. Those competitors' parent companies — Morton Salt, Honeywell, eventually ATK — were not themselves in the tire or rubber business. There was no direct parent-level competition between General Tire and Morton.

But at the technical and engineering level, the competition was fierce and personal. Engineers at Aerojet Sacramento and Hercules Bacchus Works and Thiokol Promontory competed, collaborated (on Peacekeeper stages where different companies held different stages), and circulated among employers throughout their careers. The General Tire parentage was largely irrelevant to the day-to-day technical work — which is precisely the point. The corporate structure could be as strange as the market allowed; the engineering culture was its own thing.

8.7 Further Reading

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- Jensen, Michael C. "Agency Costs of Free Cash Flow, Corporate Finance, and Takeovers." *American Economic Review* 76, no. 2 (1986): 323–329. (The theoretical argument that conglomerates destroy value — the intellectual foundation for 1980s bust-ups.)
- Hunley, J.D. *The Development of Propulsion Technology for U.S. Space-Launch Vehicles, 1926–1991.* Texas A&M University Press, 2007.
- Davis, Gerald F. *Managed by the Markets: How Finance Re-Shaped America.* Oxford University Press, 2009. (Broader context on the rise and fall of the conglomerate era.)

8.8 Exercises

1. General Tire's 1955 acquisition of RKO Pictures from Howard Hughes was one of the most unusual corporate transactions of the 1950s. Research this acquisition. What did Hughes receive, and why did he sell? What did General Tire do with the RKO assets?

2. The conglomerate model of the 1960s — acquiring unrelated businesses to diversify risk — fell out of favor in the 1980s when corporate raiders began breaking up conglomerates and returning cash to shareholders. What killed the conglomerate model? Does any version of it survive in modern corporate strategy?

3. General Tire sold its tire business to Continental AG — a German company. Research Continental AG's history. Is there irony in an American Cold War defense contractor's parent company being acquired by a German firm?

4. Map the evolution of Aerojet's ownership: General Tire → GenCorp → Aerojet Rocketdyne Holdings → L3Harris. At each transition, what changed in terms of corporate strategy, investment priorities, and management culture? What stayed the same?

Chapter 9

Strange Bedfellows: The Rest of the Family

> *"ATK was launched as an independent company in 1990, when Honeywell spun off its defense businesses to shareholders."*

Morton Salt and General Tire are the headliners, but the solid rocket industry's strange bedfellow hall of fame extends considerably further. This chapter collects the remaining remarkable partnerships — from Mason jars to thermostats, from dynamite to aluminum — that define the industry's character.

9.1 Honeywell and the Thermostat Company's Secret Defense Empire

Honeywell Incorporated was founded in Minneapolis in 1906, producing coal furnace regulators. It grew into a diversified industrial company manufacturing thermostats, controls, and building automation systems. By most accounts it was — and remains — a company associated with home comfort, climate control, and industrial instrumentation.

What most people do not know is that Honeywell spent 50 years quietly accumulating one of the most substantial defense manufacturing operations in the United States.

The defense division's roots went back to World War II, when Honeywell produced torpedo mechanisms, gun turret systems, and other military ordnance. Through the Cold War, the division expanded into propellants, ammunition, and defense electronics. By the late 1980s, Honeywell's defense businesses were generating hundreds of millions in revenue annually and employed thousands of people — almost entirely invisible to the company's public identity as a thermostat maker.

In 1990, Honeywell spun off its defense businesses to shareholders, creating **Alliant Techsystems (ATK)** as an independent company. The thermostat company had, for five decades, been secretly one of America's major defense manufacturers. ATK then proceeded to acquire Hercules Aerospace (1995) and Thiokol Propulsion (2001), assembling the dominant US solid rocket motor business from the pieces of the Cold War industrial base.

Your home thermostat company — through a chain of two corporate transactions — is the institutional ancestor of the Space Shuttle's solid rocket boosters.

9.2 DuPont → Hercules → Trident: The Antitrust-to-ICBM Pipeline

E.I. du Pont de Nemours & Company did not intend to create a solid rocket motor manufacturer. The company was forced to by the federal government in 1912, when the Department of Justice broke up DuPont's explosives monopoly and required it to spin off two independent companies.

One of those forced spin-offs — Hercules Powder Company — spent the next 80 years slowly pivoting from dynamite to rocket propellant, eventually becoming the primary supplier of solid motors for the Navy's Trident submarine-launched ballistic missile.

The causal chain: DuPont monopoly → antitrust lawsuit → forced divestiture → Hercules Powder Company → Bacchus Works, Utah → Polaris motors → Trident I → Trident II D5 → Northrop Grumman Innovation Systems.

A federal antitrust action against a gunpowder monopoly became, eight decades later, the propulsion heritage of the Navy's nuclear deterrent. No one planned this. It simply accumulated.

9.3 Ball Glass Jars → Ball Aerospace → Orbital Sciences → Pegasus

Ball Corporation was incorporated in 1880 in Buffalo, New York, producing wood-jacketed tin cans. The company moved to Muncie, Indiana in 1887 and pivoted to glass jars for food preservation — becoming the Mason jar company that has been a fixture in American kitchens since the late 19th century.

In 1956, Ball Brothers Company (as it was then known) created Ball Brothers Research Corporation — a diversified research and engineering operation that eventually became **Ball Aerospace & Technologies Corporation**. The logic was straightforward for the era: Ball had manufacturing expertise, precision engineering capabilities, and access to capital. The space age was beginning. Aerospace seemed like an attractive adjacent market.

Ball Aerospace grew into a respected manufacturer of scientific instruments, spacecraft components, and eventually complete satellites. Among its clients were NASA and the National Reconnaissance Office.

In 1989, Orbital Sciences Corporation and Hercules Aerospace co-funded the development of the Pegasus air-launched rocket. When Pegasus needed its first commercial customer to justify the private investment, **Ball Aerospace** stepped up — commissioning launches for two Ball Aerospace communications satellites before early 1991.

The Mason jar company was the anchor customer for the world's first privately developed orbital launch vehicle.

9.4 Alcoa and the One-Year Rocket Company

This story is so brief it barely registers in corporate history — yet it illustrates perfectly how industrial rationality can produce absurd outcomes.

After Morton spun off the rocket division in 1989, Thiokol Corporation renamed itself **Cordant Technologies** in 1998, acquiring Howmet International (precision turbine blade castings) and Huck International (aerospace fasteners) to diversify beyond rockets.

In 2000, Cordant Technologies merged with two divisions of Alcoa — the aluminum and materials giant — plus Howmet Castings to form **AIC Group (Alcoa Industrial Components)**. For approximately one year, the world's largest solid rocket motor manufacturer was a subsidiary of an aluminum company.

Alcoa apparently wanted only the Howmet casting business. The rocket motors were incidental. In April 2001, ATK purchased the entire operation for $2.9 billion, pulling Thiokol Propulsion out of the aluminum company's portfolio.

Duration of Alcoa's rocket ownership: approximately 12 months. Reason for ownership: they wanted turbine blade castings and the rocket company came along for the ride.

9.5 Morton Salt → Airbags → Autoliv

The Morton Salt strange bedfellow story has a coda that is almost as remarkable as the main event.

When Morton spun off Thiokol's rocket division in 1989, it retained several of Thiokol's non-rocket businesses — including an automotive airbag operation. Thiokol had been awarded the first US patent for a "safety cushion assembly for automotive vehicles" in 1953, and the company had developed airbag technology alongside its rocket work (the chemistry of rapid gas generation is relevant to both).

Morton's airbag business grew to $269 million in revenue by 1994. The salt company had become, inadvertently, one of the world's major automotive safety suppliers. It eventually sold the airbag business to Autoliv — a Swedish automotive safety company — in the late 1990s.

The complete lineage: table salt merchant (1848) → acquired a rocket company to ward off takeovers (1982) → Challenger disaster forced divestiture (1989) → retained the airbag patents → became a major airbag manufacturer → sold to a Swedish company.

9.6 Blount International: Ammunition and Rocket Boosters, Under One Roof

In 2001, ATK acquired the ammunition businesses of Blount International — including the Federal Premium, CCI, and Speer ammunition brands. This made ATK simultaneously:

- **The largest manufacturer of solid rocket motors** in the United States (via Thiokol and Hercules Aerospace)
- **The largest manufacturer of commercial ammunition** in the United States (via Federal, CCI, Speer)

In fiscal year 2001, ATK shipped the redesigned Space Shuttle solid rocket boosters and millions of rounds of .22LR, .308 Winchester, and 9mm pistol ammunition. The same corporate entity, the same management team, the same balance sheet.

The ammunition business eventually became ATK's Sporting Group, which was spun off in 2015 as **Vista Outdoor** — now a publicly traded company selling ammunition, outdoor products, and firearms accessories. Vista Outdoor's largest brands (Federal, CCI, Speer) are therefore the direct corporate descendants of ATK's 2001 ammunition acquisition, which was itself adjacent to ATK's dominant solid rocket motor business.

The sporting goods store and the missile defense contractor share a corporate grandfather.

9.7 The Pattern: Why Do Strange Bedfellows Happen?

Looking across these stories, several patterns emerge:

Chemistry is the common thread. Every bedfellow relationship has chemistry at its root. Rubber chemistry (General Tire), explosives chemistry (DuPont/Hercules), polymer chemistry (Thiokol), aluminum processing (Alcoa), gas generation (Morton/airbags). Solid rocket propulsion is fundamentally an applied chemistry problem, and the companies that could solve it were chemistry companies with expertise that transferred in unexpected directions.

Financial engineering drives the marriages. Morton needed debt. Thiokol needed stable cash flow. General Tire needed diversification. Honeywell needed to simplify its portfolio. The technical compatibility was sometimes real; the financial logic was always primary.

Defense cash flows attract non-defense capital. Steady, long-term government contracts — the kind that sustain ICBM and strategic missile programs — generate predictable free cash flow. Predictable free cash flow attracts buyers who have nothing else in common with the business. A salt company's core business is subject to commodity price swings. A rocket company with long-term government contracts is not. The economics made these marriages financially rational even when they were operationally absurd.

The divorces are always driven by crisis or strategy, rarely by success. Morton divorced Thiokol because of Challenger. Hercules sold its aerospace division because specialty chemicals were growing faster. Honeywell spun off ATK to simplify its portfolio for institutional investors. The bedfellows stay together until something forces them apart.

9.8 Further Reading

- Koistinen, Paul A.C. *Arsenal of World War II: The Political Economy of American Warfare, 1940–1945.* University Press of Kansas, 2004. (Context for how chemical and industrial companies entered defense production.)
- Markusen, Ann. *The Rise of the Gunbelt: The Military Remapping of Industrial America.* Oxford University Press, 1991. (How defense contracting reshaped American industrial geography — relevant to why these unusual parent companies ended up in Utah.)
- US Government Accountability Office. *Defense Industrial Base: Actions Needed to Improve the Availability of Critical Materials and Technologies.* GAO-23-105387, 2023.
- Orbital Sciences Corporation. *Annual Reports, 1989–2015.* (Primary source for the Orbital Sciences / Pegasus / Ball Aerospace story.)

9.9 Exercises

1. Identify the chemistry connection in each of the following pairings: Morton Salt + Thiokol, General Tire + Aerojet, Alcoa + Cordant, Honeywell + ATK. Is chemistry always the underlying rationale, or are some of these purely financial?

2. The conglomerate acquisitions described in this chapter occurred primarily in the 1960s–1990s. Modern corporate governance generally discourages this kind of unrelated diversification. What changed? Research the academic literature on "diversification discount" — the empirical finding that diversified companies trade at lower valuations than focused companies.

3. Ball Aerospace's participation as anchor customer for Pegasus was arguably critical to the program's viability. Research other cases where an unusual early customer helped make a new launch vehicle commercially viable.

4. Vista Outdoor (the sporting goods spin-off of ATK's ammunition business) is now a publicly traded company. Research its current portfolio and recent history. Does it bear any trace of its rocket company heritage?

Chapter 10

The Great Contraction: Seven to Two

> *"During the Cold War, the Pentagon bought enough solid rocket motors for intercontinental ballistic missiles to support seven suppliers. The demand for solid motors collapsed in the 1990s and dropped even further after NASA retired the space shuttle."*
> — SpaceNews, 2018

The Berlin Wall fell on November 9, 1989. Within months, the defense budget that had sustained seven solid rocket motor manufacturers began to contract. What followed was the most rapid and complete consolidation of a major American defense industrial sector since World War II.

10.1 The Peak: Seven Suppliers (circa 1980)

At the height of the Cold War, the following companies maintained significant solid rocket motor manufacturing capabilities:

1. **Thiokol** — Promontory, UT and Elkton, MD
2. **Hercules Aerospace** — Bacchus Works (Magna, UT) and Kenvil, NJ
3. **Aerojet General** — Rancho Cordova, CA
4. **Atlantic Research Corporation** — Alexandria, VA
5. **United Technologies Chemical Systems Division (CSD)** — Sunnyvale, CA
6. **Olin Aerospace** (formerly Rocket Research Corporation) — Redmond, WA
7. **Rocketdyne** (primarily liquid, with some solid capability)

Each maintained multiple facilities. Collective direct employment in solid rocket propulsion exceeded 50,000 people.

10.2 The Demand Collapse

The post-Cold War demand destruction hit on multiple fronts simultaneously:

ICBM production ended. The Minuteman production run was long over; the Peacekeeper was being cancelled. No new land-based ICBM was in development. The steady sustaining production that had kept facilities viable for decades simply stopped.

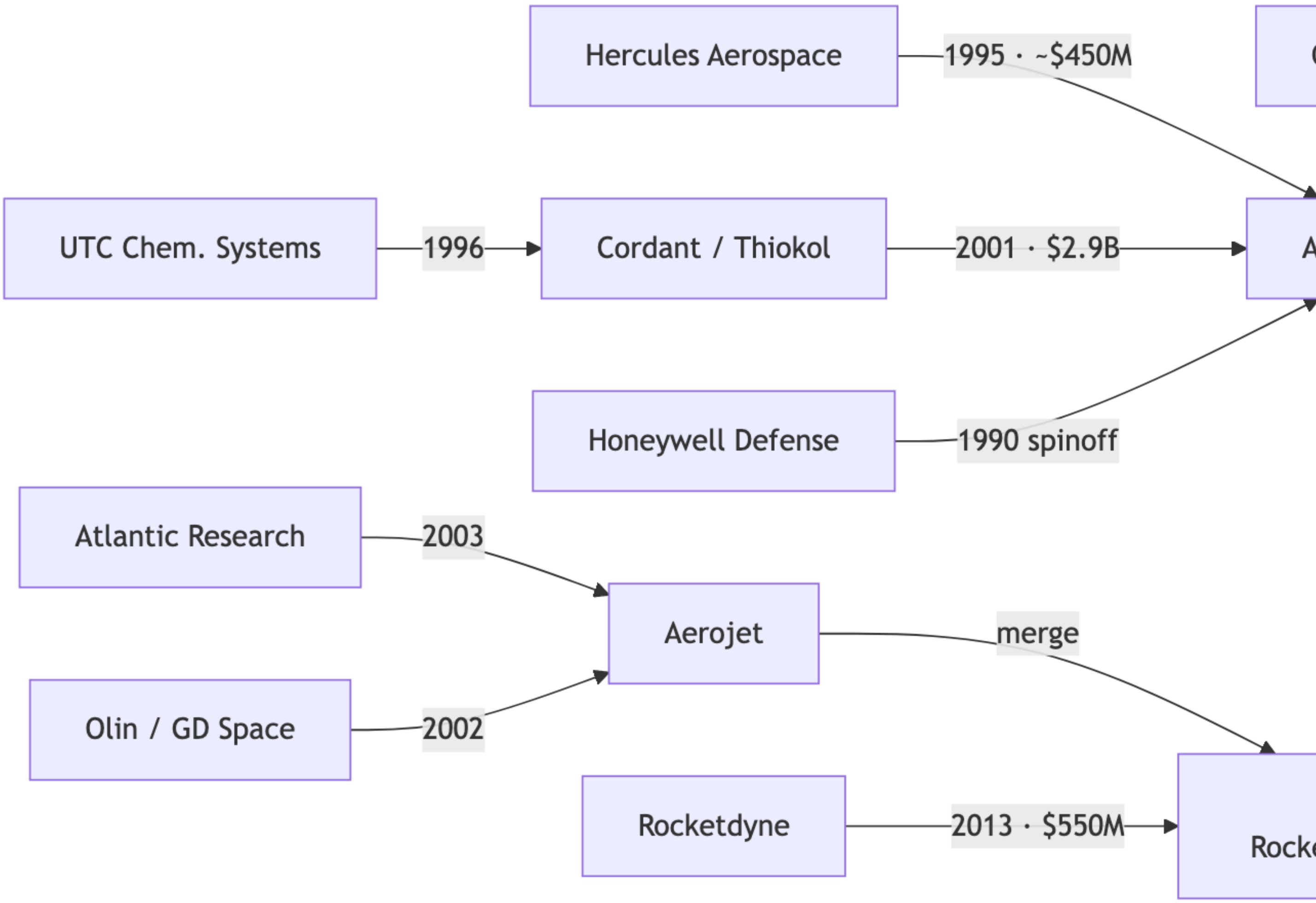

Figure 10.1: The Great Contraction: seven Cold War suppliers consolidating to two, 1990–2023. Transaction values approximate.

The Shuttle manifest shrank. NASA's planned Shuttle flight rate — which had been projected at 24 flights per year in the early 1980s — never materialized. Post-Challenger, the actual rate rarely exceeded 8 flights per year. Each flight consumed one set of Thiokol solid rocket boosters; fewer flights meant fewer motors.

Peacekeeper retirement (2005). The last MX Peacekeeper was deactivated on September 19, 2005, under the terms of the START II Treaty. Its 50 motors — representing the work of Thiokol, Aerojet, Hercules, and Rocketdyne across four stages — went to surplus. Orbital Sciences converted the Peacekeeper's motors into Minotaur launch vehicles.

Base closures. The Base Realignment and Closure (BRAC) process of the 1990s eliminated many of the government facilities — Army ammunition plants, Naval ordnance stations — that had provided sustaining work for the solid rocket industrial base.

10.3 The Consolidation Wave

The industry responded by merging — rapidly and completely.

1995: ATK acquires **Hercules Aerospace** from Hercules, Inc. for approximately $450M. The Bacchus Works joins ATK's growing portfolio. Hercules's Kenvil, NJ operations cease in 1996.

1996: United Technologies' **Chemical Systems Division** (CSD) is acquired by **Thiokol Corporation** (then renamed Cordant Technologies). CSD's Sunnyvale facilities and Trident-related work transfer to the Thiokol lineage.

1998: Thiokol Corporation renames itself **Cordant Technologies**, simultaneously acquiring Howmet International and Huck International to diversify beyond rockets.

2000: Cordant Technologies merges with Alcoa divisions to form **AIC Group**. The rocket motor business is briefly an aluminum company subsidiary.

2001: ATK purchases AIC Group's propulsion assets for $2.9 billion — also acquiring Thiokol's Promontory, UT facility and all related programs. ATK also acquires Blount International's ammunition businesses.

2002-2003: Aerojet acquires General Dynamics Space Systems' propulsion business and **Atlantic Research Corporation**. Two of the seven Cold War suppliers are absorbed into Aerojet.

2013: GenCorp acquires Pratt & Whitney Rocketdyne for $550M and merges it with Aerojet to form **Aerojet Rocketdyne**.

By 2013, seven suppliers had become two: ATK and Aerojet Rocketdyne.

10.4 The FTC's Vigilance

The Federal Trade Commission watched the consolidation carefully. When Northrop Grumman announced its acquisition of Orbital ATK (which contained ATK's solid rocket heritage) in 2017, the FTC required Northrop to supply solid rocket motors to competitors on a non-discriminatory basis as a condition of approval.

When Lockheed Martin attempted to acquire Aerojet Rocketdyne in 2020–2022, the FTC blocked it entirely on the grounds that it would eliminate the only independent solid rocket motor manufacturer and give Lockheed vertical integration advantages over competitors. This was the most aggressive FTC action in the solid rocket industry since the original Sherman Act case against DuPont in 1912.

10.5 The Human Cost

The consolidation eliminated tens of thousands of high-skill, high-wage jobs in specific communities that had no equivalent employers. Promontory, Utah — essentially a company town built around the Thiokol facility — survived because ATK and then Northrop continued operations. Kenvil, New Jersey did not survive — the facility closed in 1996, ending 80 years of energetic materials manufacturing.

Sacramento's Aerojet complex — which had employed over 20,000 people during the 1960s space race — was reduced to a fraction of that workforce by the 2010s, with manufacturing operations being consolidated away from the original site.

The industrial towns built around these facilities — Magna, Utah; Kenvil, New Jersey; portions of Sacramento County — bear the marks of this contraction in their economic statistics to this day.

10.6 What Survived: The Two Survivors

Northrop Grumman Innovation Systems — containing the Thiokol, Hercules, and Orbital Sciences heritage — holds the dominant position. Its programs include Trident II D5 (all three stages), Minuteman III sustaining, SLS five-segment solid rocket boosters, Ground-Based Midcourse Defense interceptor motors, and the emerging LGM-35A Sentinel (Minuteman replacement).

L3Harris / Aerojet Rocketdyne — containing the Aerojet, Atlantic Research, CSD, and Rocketdyne heritage — holds the second position. Its programs include Ground-Based Interceptor (some programs), RS-25 engine (SLS liquid), RL-10 engine (Centaur upper stage), and various tactical missile motors.

The FTC's requirement that Northrop supply motors to competitors on a non-discriminatory basis is the only structural protection preventing this from becoming a functional monopoly.

10.7 Further Reading

- Markusen, Ann, and Sean Costigan (eds.). *Arming the Future: A Defense Industry for the 21st Century.* Council on Foreign Relations Press, 1999. (Policy perspectives on post-Cold War defense industrial consolidation.)
- Gholz, Eugene, and Harvey M. Sapolsky. "Restructuring the US Defense Industry." *International Security* 24, no. 3 (1999–2000): 5–51. (Academic analysis of why consolidation occurred and whether two suppliers are sufficient.)
- United States General Accounting Office. *Defense Industrial Base: Trends in DoD Spending, Industrial Productivity, and Competition.* GAO/PEMD-97-3, 1997.

- Bitzinger, Richard A. "The Globalization of the Arms Industry: The Next Proliferation Challenge." *International Security* 19, no. 2 (1994): 170–198.

10.8 Exercises

1. Map the 1990–2013 consolidation as a timeline showing each acquisition, the acquiring entity, the acquired entity, and the approximate transaction value. What is the total capital deployed in the consolidation? Who were the winners and losers financially?
2. The BRAC process eliminated government facilities that had provided sustaining work for the solid rocket industrial base. Research the BRAC process. How does the government decide which facilities to close, and how were communities expected to adapt?
3. The FTC blocked Lockheed's acquisition of Aerojet Rocketdyne but allowed Northrop's acquisition of Orbital ATK (with conditions). What accounts for the different outcomes? Research the economic arguments the FTC used in each case.
4. Design a "counterfactual" industrial structure: if the Cold War had not ended in 1989, and demand had remained at 1985 levels through 2010, how many suppliers would likely have survived? What would the industry look like today?

Chapter 11

The ATK Era: When One Company Ate the Industry

> *"With the Thiokol acquisition, ATK became the dominant supplier of solid-fuel rocket motors."* — Utah History Quarterly

Alliant Techsystems — ATK — is the company that consolidated the American solid rocket industry. Between 1990 and 2015, it assembled the Thiokol, Hercules, and supporting propulsion heritage under a single roof, becoming the largest solid rocket motor manufacturer in the world and, for a time, the largest ammunition manufacturer in the United States simultaneously.

The story of ATK is the story of one company making a series of acquisitions so consequential that they permanently restructured a strategic industry.

11.1 The Honeywell Spinoff (1990)

ATK was launched as an independent company in 1990, when Honeywell spun off its defense businesses to shareholders. The former Honeywell businesses had supplied defense products and systems to the US and its allies for 50 years. The initial ATK was primarily a propellants, ammunition, and defense systems manufacturer — significant, but not yet dominant in solid rocket motors specifically.

The transformation came through two acquisitions.

11.2 Hercules Aerospace: The First Acquisition (1995)

In 1995, ATK purchased **Hercules Aerospace Company** from Hercules, Inc. for approximately $450 million. The acquisition brought:

- The **Bacchus Works** in Magna, Utah — the 1913 dynamite plant that had become a premier solid rocket facility
- The **Trident II D5** propulsion program — all stages for the Navy's primary submarine-launched ballistic missile
- The **Pegasus** rocket motor production — the Orion 50, 50XL, and 38 stages
- The **Minuteman III Stage 3** heritage and any sustaining work

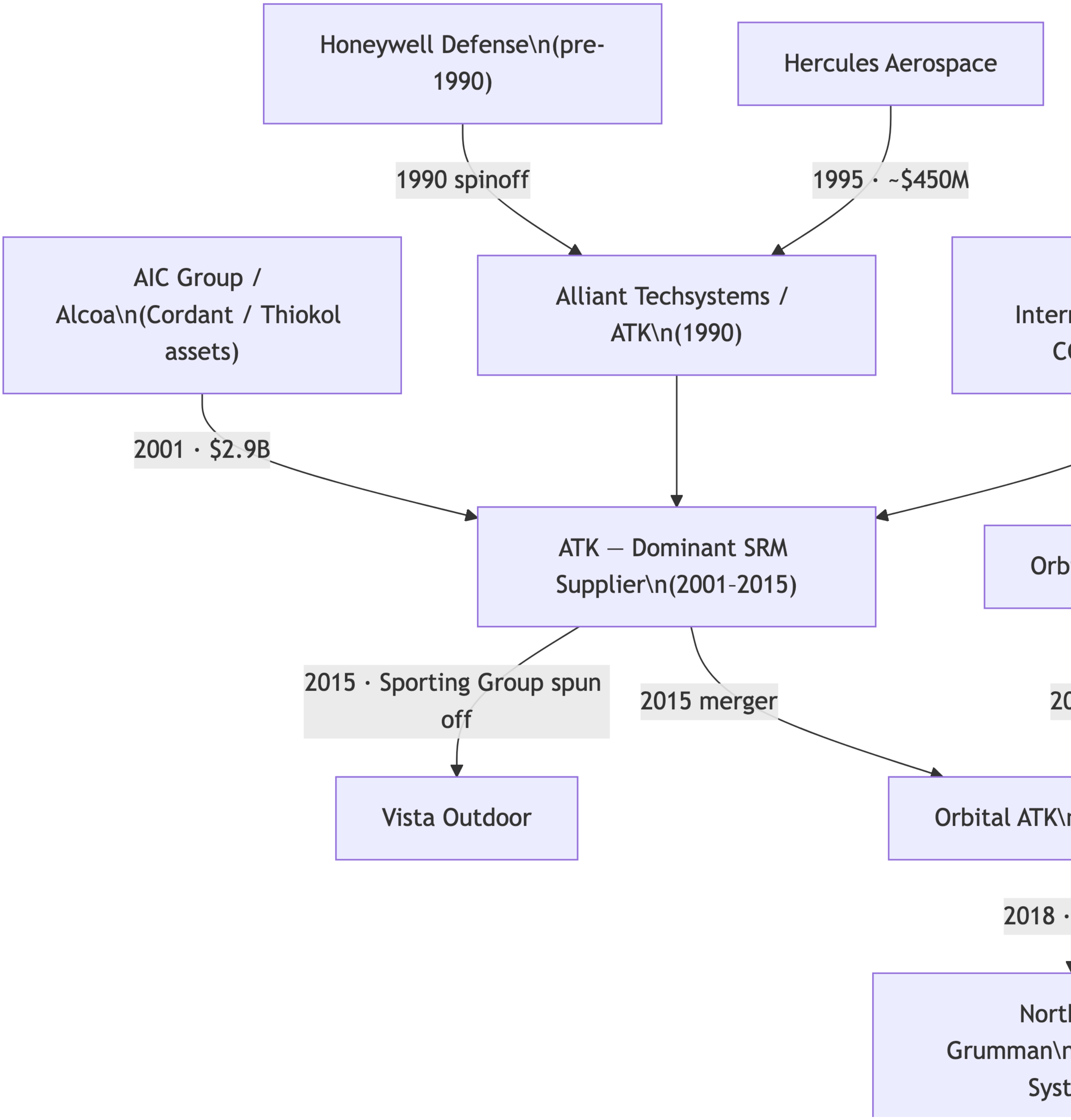

Figure 11.1: ATK's consolidation of the American solid rocket motor industry, 1990–2018.

- Filament-wound graphite composite casing technology
- Approximately 4,500 skilled workers in Utah

ATK already had a presence in Utah through its existing operations. The Hercules acquisition transformed it from a presence into a dominant force. The Bacchus Works — after operating under six corporate owners since 1913 — was now part of ATK.

11.3 Cordant/Thiokol: The Decisive Acquisition (2001)

The Thiokol acquisition was the larger and more consequential move. The path was indirect:

1. Thiokol renamed itself **Cordant Technologies** in 1998, acquiring Howmet International and Huck International
2. In 2000, **Alcoa** acquired Cordant, forming AIC Group — wanting primarily the Howmet casting business
3. In April 2001, ATK purchased the propulsion assets from AIC/Alcoa for **$2.9 billion**

The $2.9 billion figure was the largest acquisition in ATK's history and one of the largest transactions in the solid rocket industry. It brought:

- **Thiokol Propulsion** — the primary manufacturer of the Space Shuttle solid rocket boosters
- The **Promontory, Utah** facility — over 10,000 acres of test infrastructure
- **Minuteman III Stage 1** heritage
- **Star 37 and Star 48** upper stage motors — the workhorses of interplanetary spacecraft separation
- The Shuttle SRB program sustaining work

With the Thiokol acquisition, ATK controlled the lion's share of the US solid rocket fuel market. ATK built the third stage of the Trident missile [via Hercules] and had earlier bought Hercules Aerospace Co., builder of the second stage. With Thiokol, ATK now owned first stage heritage as well.

11.4 Ammunition: The Unexpected Expansion

Also in 2001, ATK acquired the ammunition businesses of Blount International — including Federal Premium, CCI, and Speer brands. This made ATK the nation's largest manufacturer of ammunition. The decision to combine a dominant solid rocket motor business with a dominant commercial ammunition business in one company was unusual even by the standards of the defense industry.

The rationale: both businesses involved energetic materials (propellants and propellant chemistry), both involved precision manufacturing, and both involved long-term government contracts alongside commercial sales. ATK's propellant expertise was genuinely applicable across both product lines.

The ammunition business eventually became ATK's Sporting Group and was spun off in 2015 as Vista Outdoor — a publicly traded company that continues to operate the Federal, CCI, and Speer brands.

11.5 The Shuttle Era Programs

During the ATK era, the company was the sole manufacturer of the reusable Solid Rocket Motor used to launch the NASA Space Shuttle. After the Shuttle program ended in 2011, ATK developed the five-segment solid rocket booster for NASA's Space Launch System — a significant redesign and upscale of the four-segment Shuttle SRB.

ATK also produced the launch abort motor for NASA's Orion capsule — a small but critical solid motor that would pull the capsule away from an exploding rocket during a launch abort.

11.6 The Merger: Orbital ATK (2014–2015)

On April 29, 2014, ATK announced its board had unanimously approved a plan to split the company and merge its aerospace and defense operations with **Orbital Sciences Corporation**.

The structure: - ATK's **Sporting Group** was spun off to ATK shareholders as **Vista Outdoor** (tax-free) - ATK's **Aerospace and Defense Groups** merged with Orbital Sciences to form **Orbital ATK** - Orbital Sciences shareholders received ATK common stock

The merger closed February 9, 2015. Orbital ATK began operations February 20, 2015.

The merger made strategic sense: Orbital Sciences had launch vehicle capabilities (Pegasus, Taurus, Minotaur, Antares) and satellite manufacturing; ATK had dominant solid rocket motor production. Together, they formed a vertically integrated aerospace and defense company — the motors and the vehicles that used them under one roof.

Northrop Grumman acquired Orbital ATK in June 2018 for $7.8 billion — bringing the entire consolidated ATK lineage (Thiokol heritage, Hercules heritage, Orbital Sciences launch vehicles) into the second-largest defense contractor in the United States.

11.7 The Legacy

ATK's 25-year consolidation (1990–2015) permanently restructured the American solid rocket industry. By the time Northrop absorbed Orbital ATK, the seven Cold War suppliers had been reduced to effectively one dominant player (Northrop/ATK heritage) and one competitor (Aerojet Rocketdyne). The consolidation was complete.

The Promontory facility — once Thiokol, then Morton Thiokol, then Thiokol Corporation, then Cordant, then AIC Group, then ATK Thiokol, then ATK Launch Systems Group, then Orbital ATK — now operates as **Northrop Grumman Promontory**. The name on the gate changes. The test stands remain.

11.8 Further Reading

- Utah History Quarterly. "Alliant Techsystems and the Utah Aerospace Industry." Various issues.
- Defense Science Board. *Task Force Report: Preserving a Healthy and Competitive US Defense Industry to Ensure our National Security.* Office of the Under Secretary of Defense for Acquisition, Technology and Logistics, 2011.

- Orbital Sciences Corporation. *Annual Reports and SEC Filings, 1990–2015*. (Primary source for the Orbital Sciences side of the merger.)
- US Senate Armed Services Committee. *Hearings on the State of the Defense Industrial Base*. Various years, 2010–2018. (Congressional perspective on ATK's consolidation and its implications for competition.)

11.9 Exercises

1. ATK paid $2.9 billion for Cordant/Thiokol in 2001. Research what that implies about Thiokol's valuation metrics at the time. Was this a fair price for the seller? For the buyer?
2. The decision to combine solid rocket motors and commercial ammunition in one company (via the Blount acquisition) was unusual. Construct the pro and con arguments for this combination. Which would you find more persuasive as an ATK board member in 2001?
3. The Orbital ATK merger was structured to spin off the Sporting Group to ATK shareholders while merging the aerospace business with Orbital Sciences. Research the tax and structural logic of this transaction. Why was it structured this way rather than as a simple merger?
4. Northrop Grumman paid $7.8 billion for Orbital ATK in 2018. Research what programs Northrop was acquiring and evaluate whether this was a fair price given the backlog, programs, and strategic value of the combined entity.

Chapter 12

Two Survivors: The Duopoly and What Comes Next

> *"There are now technically two companies that still manufacture large solid rockets for military ICBMs — Aerojet Rocketdyne and Northrop Grumman Innovation Systems."*
> — SpaceNews, 2018

Seven Cold War suppliers. Fifty years of mergers, acquisitions, spinoffs, disasters, and consolidation. Two survivors.

This is where the family tree ends — for now.

12.1 The Duopoly

Northrop Grumman Innovation Systems (NGIS), created when Northrop Grumman acquired Orbital ATK in June 2018 for $7.8 billion, is the dominant player. NGIS contains the propulsion heritage of Thiokol, Hercules, and Orbital Sciences — the three largest Cold War solid rocket lineages — plus Northrop's own legacy propulsion work.

Current NGIS programs include: - **Trident II D5** — all three stages of the Navy's primary nuclear deterrent - **Minuteman III** — sustaining production and heritage programs - **LGM-35A Sentinel** — the Minuteman III replacement, the most significant new ICBM program in decades - **SLS Five-Segment SRB** — the solid rocket boosters for NASA's Space Launch System - **Ground-Based Midcourse Defense** — interceptor motors for missile defense - **Pegasus XL** — the air-launched rocket (final flights being conducted) - **Antares / Cygnus** — ISS resupply missions - **Orion Launch Abort Motor and Attitude Control Motor**

Northrop Grumman has 100-percent propulsion success on strategic production motors — a record the company prominently advertises, with good reason.

L3Harris / Aerojet Rocketdyne, formed when L3Harris Technologies acquired Aerojet Rocketdyne in July 2023 for $4.7 billion, is the competitor. Aerojet Rocketdyne contains the propulsion heritage of Aerojet, Atlantic Research Corporation, General Dynamics Space Systems, and Rocketdyne.

Current Aerojet Rocketdyne programs include: - **RS-25 engine** — the main engines for NASA's

Space Launch System - **RL-10 engine** — the workhorse upper-stage engine used on Centaur, SLS upper stage - **Ground-Based Interceptor** — solid motors for missile defense - **Sentinel ICBM** — competing for propulsion contracts against Northrop - **AJ-60A** — strap-on solid boosters for Atlas V - Various tactical missile motors

12.2 The FTC's Role

The Federal Trade Commission has been the structural architect of this duopoly — both by allowing the consolidation that created it and by setting limits on further consolidation.

The Northrop-Orbital ATK approval (2018): The FTC approved the acquisition but required Northrop to supply solid rocket motors to competitors on a non-discriminatory basis. This was designed to prevent Northrop from using its SRM dominance to disadvantage competitors in launch vehicle and missile programs.

The Lockheed-Aerojet Rocketdyne rejection (2022): The FTC voted 4-0 to block Lockheed Martin's $4.4 billion acquisition of Aerojet Rocketdyne, ruling that Lockheed's vertical integration would disadvantage other defense contractors who depended on Aerojet for propulsion. Lockheed abandoned the deal in February 2022.

The L3Harris acquisition (2023): When L3Harris acquired Aerojet Rocketdyne for $4.7 billion, the FTC approved the deal. L3Harris — primarily a defense electronics and communications company — was not seen as creating the same vertical integration concerns as Lockheed, since L3Harris does not itself manufacture missiles, spacecraft, or launch vehicles that would directly benefit from preferential propulsion access.

The FTC's distinction between the Lockheed rejection and the L3Harris approval illustrates the regulatory logic: it is not the size of the transaction that matters, but whether the acquirer competes with the customers who depend on the target's products.

12.3 The Sentinel: First New ICBM in Decades

The LGM-35A Sentinel program — the Air Force's replacement for the aging Minuteman III — is the most consequential solid rocket motor program in decades. Northrop Grumman is the prime contractor for the missile itself and holds the propulsion contract. Aerojet Rocketdyne is competing for portions of the propulsion work.

The Sentinel's development has been troubled. In January 2024, the Air Force disclosed that the program's cost had grown by more than 37%, triggering a Nunn-McCurdy breach requiring Congressional notification and program review. The cost growth reflects both the difficulty of developing a new ICBM from scratch and the challenges of maintaining an industrial base that has contracted significantly since the Minuteman was developed in the 1960s.

The institutional knowledge that designed Minuteman — the engineers who understood solid propellant grain geometry, case design, nozzle dynamics at operational scale — has largely retired or died. Rebuilding that capability is one of the defining challenges of the Sentinel program.

12.4 New Space: The Third Category

The duopoly describes the strategic/large solid rocket motor market. Below it, a new category of solid rocket motor manufacturer is emerging in the New Space era:

Rocket Lab — the New Zealand/American launch company, produces solid kick stages (Curie, Photon) and is developing larger solid motor components.

Firehawk Aerospace — a Texas-based startup developing hybrid and solid motors for tactical and commercial applications.

Evolution Space — focused on small tactical solid motors.

SpinLaunch — though not a traditional SRM company, represents alternative propulsion approaches.

None of these New Space companies currently competes with Northrop or L3Harris/Aerojet Rocketdyne for strategic (nuclear deterrent) solid rocket motors. The barriers to entry in that segment — the technical requirements, the security clearances, the regulatory frameworks, the testing infrastructure — remain formidable.

12.5 The Open Question: Is Two Enough?

Defense analysts have debated whether a duopoly is sufficient for the strategic solid rocket motor industrial base. The concern is not primarily about price — the government is the only customer, and it can regulate pricing — but about resilience. If one supplier suffers a significant accident, a facility fire, a quality escape, or a production disruption, is there sufficient capacity and capability at the other to maintain deterrence?

The Challenger O-ring failure demonstrated that a single production problem at a single company can ground the entire US space launch fleet. The Minuteman III force currently depends on aging motors that are increasingly difficult to maintain. The Sentinel program is behind schedule and over budget.

These are not merely industrial policy questions. They are questions about whether the United States maintains the capacity to build and sustain the weapons systems it has decided are essential to its national security — and whether the consolidation that made business sense in the 1990s has created strategic vulnerabilities it was too early to see.

12.6 A Final Word on the Incest

The industry that set out to design and manufacture the most destructive weapons in human history ended up being run by salt merchants, tire makers, thermostat companies, aluminum producers, and jar makers. The engineers who actually built the motors — who understood propellant chemistry, grain design, nozzle erosion, case structural margins — persisted through every corporate transformation, carrying institutional knowledge that no merger agreement could transfer or no acquisition price could fully capture.

That is what “incestuous community” really means. Not that the corporate names are familiar, but that the people are. The engineer who started at Thiokol in 1965, transferred to ATK in

2001, retired from Northrop Grumman in 2018 — their career is the continuity that the family tree represents in organizational charts but cannot fully capture.

The corporations come and go. The knowledge stays.

12.7 Further Reading

- Gholz, Eugene, and Harvey M. Sapolsky. "Restructuring the US Defense Industry." *International Security* 24, no. 3 (1999–2000): 5–51.
- Congressional Budget Office. *Approaches to Reducing Federal Spending on Military Hardware*. CBO, 2021. (Includes analysis of sole-source strategic propulsion procurement costs.)
- Asker, James R. "The Solid Rocket Duopoly." *Aviation Week & Space Technology*, March 2019.
- US Air Force. *LGM-35A Sentinel Nunn-McCurdy Breach Certification Report*. Department of the Air Force, 2024. (Primary source for the Sentinel cost-growth crisis.)
- Loren Thompson. "Why the US Can't Afford to Lose Its Second Solid-Rocket Motor Maker." *Forbes / Lexington Institute*, February 2022. (Industry perspective on the FTC's Lockheed-Aerojet decision.)

12.8 Exercises

1. The FTC approved the L3Harris acquisition of Aerojet Rocketdyne but blocked the Lockheed acquisition. Construct the economic argument for each decision. Do you find the FTC's reasoning consistent?

2. The Sentinel ICBM program experienced a 37%+ cost growth, triggering a Nunn-McCurdy breach. Research the Nunn-McCurdy Act. What does it require when cost growth exceeds thresholds? What are the likely outcomes for the Sentinel program?

3. The industrial base for strategic solid rocket motors has contracted from seven suppliers to two. Construct a national security argument for maintaining more than two suppliers. Then construct the counter-argument that two suppliers are sufficient. Which do you find more persuasive?

4. This book began with the premise that the solid rocket industry is an "incestuous community." Having read the lineages, the strange bedfellows, and the consolidation history — do you agree? Is the incest primarily corporate (entities merging and splitting), human (engineers moving between companies), or technical (propellant chemistry and design knowledge transferring)?

References

Appendix A: Corporate Genealogy Reference

This appendix provides a consolidated reference for every significant corporate transaction in the American solid rocket motor industry from 1912 to the present. Events are organized by current inheritor, then chronologically.

Northrop Grumman Innovation Systems (NGIS) Heritage

NGIS contains the merged heritage of Thiokol, Hercules, UTC Chemical Systems Division, and Orbital Sciences.

Thiokol / Cordant / ATK Lineage

Year	Event	From	To	Approx. Value
1929	Founded	—	Thiokol Chemical Corporation	—
1982	Acquisition	Morton International	Morton Thiokol, Inc.	~$200M
1989	Spinoff	Morton Thiokol	Thiokol Corporation (independent)	—
1996	Acquisition	United Technologies (CSD)	Thiokol Corporation / Cordant	not disclosed
1998	Rename	Thiokol Corporation	Cordant Technologies	—

Year	Event	From	To	Approx. Value
2000	Merger	Cordant Technologies + Alcoa divisions	AIC Group (Alcoa Industrial Components)	—
2001	Acquisition of propulsion assets	AIC Group / Alcoa	Alliant Techsystems (ATK)	$2.9B

Hercules Lineage

Year	Event	From	To	Approx. Value
1912	Federal antitrust divestiture	DuPont (ordered by court)	Hercules Powder Company	—
1966	Rename	Hercules Powder Company	Hercules, Inc.	—
1978	Division created	Hercules, Inc.	Hercules Aerospace Company (division)	—
1995	Acquisition of aerospace division	Hercules, Inc.	Alliant Techsystems (ATK)	~$450M

UTC Chemical Systems Division Lineage

Year	Event	From	To	Approx. Value
1960s	Founded as division	United Technologies Corp.	Chemical Systems Division (CSD), Sunnyvale CA	—

Year	Event	From	To	Approx. Value
1996	Acquisition	United Technologies Corp.	Thiokol Corporation / Cordant Technologies	not disclosed

Orbital Sciences Lineage

Year	Event	From	To	Approx. Value
1982	Founded	—	Orbital Sciences Corporation	—
1990	Honeywell spinoff	Honeywell Inc.	Alliant Techsystems (ATK)	—
2001	Blount ammunition acquisition	Blount International	ATK (Sporting Group)	~$200M
2015	Merger	ATK Aerospace + Orbital Sciences Corporation	Orbital ATK	—
2015	Spinoff	ATK	Vista Outdoor (Sporting Group)	—
2018	Acquisition	Orbital ATK	Northrop Grumman Innovation Systems	$7.8B

L3Harris / Aerojet Rocketdyne Heritage

Aerojet Rocketdyne contains the merged heritage of Aerojet, Atlantic Research Corporation, Olin/Primex/General Dynamics Space Systems, and Rocketdyne.

Aerojet Lineage

Year	Event	From	To	Approx. Value
1942	Founded	GALCIT alumni + General Tire investment	Aerojet Engineering Corporation	—
1942	Major investment	General Tire & Rubber Company	Aerojet (majority shareholder)	—
1984	Rename / restructure	General Tire & Rubber Company	GenCorp, Inc. (holding company)	—
2002	Acquisition	General Dynamics (Space Propulsion business, ex-Olin/Primex)	Aerojet	not disclosed
2003	Acquisition	Atlantic Research Corporation	Aerojet	not disclosed
2013	Acquisition	Pratt & Whitney Rocketdyne (United Technologies)	GenCorp / Aerojet	$550M
2013	Merger	Aerojet + Rocketdyne	Aerojet Rocketdyne	—
2015	Rename	GenCorp, Inc.	Aerojet Rocketdyne Holdings, Inc.	—
2022	Blocked acquisition	Lockheed Martin attempted to acquire Aerojet RD	FTC blocked 4-0; Lockheed withdrew	$4.4B

Year	Event	From	To	Approx. Value
2023	Acquisition	Aerojet Rocketdyne Holdings	L3Harris Technologies	$4.7B

Atlantic Research Corporation (ARC) Lineage

Year	Event	From	To	Approx. Value
1950	Founded	—	Atlantic Research Corporation, Alexandria VA	—
1994	Acquisition	ARC	Sequa Corporation	not disclosed
2003	Acquisition	Sequa Corporation (ARC)	Aerojet	not disclosed

Olin / Primex / General Dynamics Space Systems Lineage

Year	Event	From	To	Approx. Value
~1960s	Founded	—	Rocket Research Corporation, Redmond WA	—
~1985	Acquisition	Olin Corporation	Olin Aerospace Company (formerly Rocket Research)	—
1996	Acquisition	Olin Corporation	Primex Technologies	not disclosed

Year	Event	From	To	Approx. Value
2001	Acquisition	Primex Technologies	General Dynamics Ordnance and Tactical Systems	not disclosed
2002	Acquisition of Space Propulsion business	General Dynamics	Aerojet	not disclosed

Rocketdyne Lineage

Year	Event	From	To	Approx. Value
1955	Rocketdyne Division established	North American Aviation	Rocketdyne Division	—
1967	Merger	North American Aviation + Rockwell-Standard	North American Rockwell / Rockwell International	—
1996	Acquisition	Rockwell International (Rocketdyne Division)	Boeing Company	—
2005	Acquisition	Boeing (Rocketdyne)	United Technologies / Pratt & Whitney	$700M
2013	Acquisition	Pratt & Whitney Rocketdyne	GenCorp / Aerojet	$550M

Regulatory and Antitrust Events

Year	Case / Event	Parties	Outcome
1912	*United States v. E.I. du Pont de Nemours* (Sherman Act)	DOJ vs. DuPont	DuPont ordered to divest; created Hercules Powder Co. and Atlas Powder Co.
2018	FTC review: Northrop Grumman / Orbital ATK	FTC, Northrop, Orbital ATK	Approved with behavioral remedy: Northrop required to supply SRMs to competitors on non-discriminatory basis
2022	FTC vs. Lockheed Martin / Aerojet Rocketdyne	FTC, Lockheed, Aerojet RD	Blocked 4-0; Lockheed withdrew February 2022
2023	FTC review: L3Harris / Aerojet Rocketdyne	FTC, L3Harris, Aerojet RD	Approved; L3Harris not a vertical integration concern

Key Facilities: Current Status

Facility	Location	Original Owner	Current Operator	Active Since
Promontory	Box Elder County, UT	Thiokol Chemical (1957)	Northrop Grumman	1957
Bacchus Works	Magna, UT	Hercules Powder (1913)	Northrop Grumman	1913
Elkton, MD	Cecil County, MD	Thiokol Chemical (1951)	Northrop Grumman	1951
Sacramento Complex	Rancho Cordova, CA	Aerojet (1950)	L3Harris / Aerojet RD (reduced)	1950
Camden, AR	Ouachita County, AR	Aerojet / Atlantic Research	L3Harris / Aerojet RD	1960s

Program-to-Producer Reference

Program	Stage	Cold War Producer	Current Operator
Minuteman III	Stage 1	Thiokol	Northrop Grumman (sustainment)

Program	Stage	Cold War Producer	Current Operator
Minuteman III	Stage 2	Aerojet	L3Harris / Aerojet RD (sustainment)
Minuteman III	Stage 3	Hercules	Northrop Grumman (sustainment)
Trident II D5	Stages 1–3	Hercules (primary), Thiokol	Northrop Grumman
Space Shuttle SRB	Single (4-segment)	Morton Thiokol / Thiokol / ATK	Program ended 2011
SLS SRB	Single (5-segment)	ATK / Orbital ATK	Northrop Grumman
Peacekeeper (MX)	Stage 1	Thiokol	Retired 2005
Peacekeeper (MX)	Stage 2	Aerojet	Retired 2005
Peacekeeper (MX)	Stage 3	Hercules	Retired 2005
Peacekeeper (MX)	Stage 4	Rocketdyne	Retired 2005
Pegasus XL	Stages 1–3	Hercules / ATK	Northrop Grumman (final flights)
LGM-35A Sentinel	TBD	—	Northrop Grumman (prime)

Appendix B: Glossary

Technical and organizational terms used throughout this book.

Propellant Chemistry and Motor Design

Ammonium Perchlorate (AP) The oxidizer used in virtually all modern US composite solid propellants. AP provides the oxygen required for propellant combustion. A typical formulation contains 65–70% AP by weight. Manufactured primarily at a single US facility (Kerr-McGee / successor, now Northrop Grumman's HSMC in Henderson, NV), making AP supply a recognized single-point vulnerability in the defense industrial base.

APCP (Ammonium Perchlorate Composite Propellant) The dominant class of modern solid rocket propellant. Consists of AP oxidizer particles, a polymeric fuel-binder (typically HTPB), aluminum powder fuel, and curing agents, cast as a homogeneous solid. Burns by deflagration rather than detonation. Enabled the large-grain motors required by ICBM and space launch applications.

Burn Rate The rate at which the propellant surface recedes during combustion, typically expressed in inches per second at a reference pressure. Burn rate determines thrust level and duration. Controlled through propellant formulation (AP particle size, additives) and grain geometry. Increases with chamber pressure — described by Saint-Robert's law: $r = aP\hat{}n$.

Case Mass Fraction The ratio of propellant mass to total motor mass (propellant + case + nozzle). Higher case mass fraction means more propellant per unit of total motor weight, improving vehicle performance. The drive toward higher case mass fraction motivated the transition from steel to aluminum to filament-wound composite cases.

Composite Propellant A solid propellant consisting of solid oxidizer particles (AP) embedded in a continuous polymeric fuel-binder matrix. Contrasted with double-base propellants. Composite propellants can be cast in large grain sizes with controlled burn characteristics, enabling the large motors required for ICBMs and large launch vehicles.

Deflagration Rapid subsonic combustion propagating by heat transfer from the burning surface into the unburned propellant. What solid rocket propellants are designed to do. Contrasted with detonation (supersonic shock-wave propagation), which is catastrophic and must be prevented. The distinction between a rocket motor and a bomb is that the propellant deflagrates rather than detonates.

Double-Base Propellant A solid propellant consisting of nitrocellulose and nitroglycerin —

both the fuel and oxidizer are combined within the same molecule. Inherited from the smokeless powder industry. Used extensively in WWII rockets (bazooka, JATO), but limited in scale and energy density compared to composite propellants.

Field Joint The connection between propellant segments in a segmented solid rocket motor case. In the Space Shuttle SRBs, four propellant segments were manufactured at Promontory, UT and joined at Kennedy Space Center using field joints sealed with O-rings. The failure of an O-ring at the aft field joint of the right SRB caused the Challenger accident on January 28, 1986.

Grain The shaped propellant charge inside a solid rocket motor case. The grain geometry — the three-dimensional shape of the propellant and its internal void — determines the surface area exposed to combustion at any point in the burn, and therefore the thrust-time profile. Common grain geometries include cylindrical core, star perforation, finocyl (fin + cylinder), and wagon wheel.

Grain Geometry The cross-sectional profile of a solid propellant grain, designed to produce a desired thrust-time profile. A **star perforation** (internal star-shaped cavity) produces a roughly neutral (constant) burn because the star tips burn away at the same rate new surface is exposed in the valley regions. A simple cylindrical core produces a progressive (increasing) burn. Grain geometry design is a core discipline of solid rocket motor engineering.

HTPB (Hydroxyl-Terminated Polybutadiene) The fuel-binder polymer used in most modern US composite solid propellants, including the Shuttle SRBs and SLS boosters. Replaced earlier polyurethane and polysulfide binders beginning in the 1970s. HTPB offers better mechanical properties at low temperatures, higher energy density, and better processing characteristics than its predecessors.

Internal Ballistics The discipline governing combustion, gas dynamics, and motor performance inside a solid rocket motor case. Internal ballistics models predict chamber pressure, thrust, and burn time from propellant properties and grain geometry. A technically demanding field — the same motor, fired at different temperatures or altitudes, will produce different performance.

JATO (Jet-Assisted Take-Off) Solid or liquid rocket units attached to an aircraft to supplement engine thrust during takeoff from short runways or carrier decks. The development of JATO units at Caltech's GALCIT in the late 1930s was the direct origin of the American solid rocket industry.

Nozzle Erosion Degradation of the nozzle throat material during firing caused by the hot, high-velocity combustion gases. The throat diameter increases as it erodes, changing the motor's pressure and thrust characteristics. Managing nozzle erosion — through material selection (graphite, carbon-carbon, ablative coatings) and motor design — is a persistent engineering challenge.

O-Ring An elastomeric seal used at field joints and other interfaces to prevent hot combustion gas from escaping the motor case. In the Shuttle SRBs, the primary and secondary O-rings at each field joint were the only barrier between combustion gases and the exterior. The O-ring material becomes less flexible at cold temperatures, reducing its ability to seal properly — the mechanism behind the Challenger failure.

Polysulfide Rubber The material Thiokol Chemical Corporation was originally founded to produce. A synthetic rubber with exceptional chemical and solvent resistance, developed accidentally by Joseph Patrick in 1929. Thiokol's polysulfide polymers were the original binder material for composite solid propellants in the early 1950s, before being replaced by polyurethane and eventually HTPB formulations.

Propellant Ballistics Commonly used as a synonym for internal ballistics; specifically the study of how propellant properties (burn rate, pressure exponent, temperature sensitivity) translate into motor performance.

Specific Impulse (Isp) The primary efficiency metric for rocket propellants, defined as thrust produced per unit propellant weight flow (units: seconds). Higher Isp means more thrust per pound of propellant consumed. Composite propellants with aluminum fuel achieve Isp values of approximately 250–270 seconds in vacuum. Liquid propellant engines typically achieve higher Isp (300–450 seconds), which is why upper stages often use liquid propulsion.

Star Perforation A grain geometry in which the internal void is star-shaped. As the propellant burns outward from the star tips and valleys, the total burning surface area remains approximately constant — producing a neutral (flat) thrust-time profile. Widely used in ICBM stages where constant thrust is desired.

Temperature Sensitivity The dependence of propellant burn rate on initial temperature. A motor fired at hot conditions (e.g., a missile stored in a desert) burns faster and at higher pressure than one fired at cold conditions. Temperature sensitivity is characterized by the coefficient _p (pressure sensitivity) and _r (burn rate sensitivity). Managing temperature sensitivity across the operational temperature range (typically −65°F to +165°F for US strategic systems) is a key design constraint.

Programs and Systems

ICBM (Intercontinental Ballistic Missile) A ballistic missile with a range exceeding 5,500 km, capable of delivering nuclear warheads between continents. Current US ICBMs: Minuteman III (land-based, in service). Replacement: LGM-35A Sentinel (in development).

LGM-35A Sentinel The US Air Force program to replace the Minuteman III ICBM. Northrop Grumman is the prime contractor. As of 2024, the program has experienced a Nunn-McCurdy cost breach (>37% growth over baseline), triggering Congressional review. The program represents the first new US ICBM development in decades.

Minuteman III The US Air Force's land-based ICBM, deployed since 1970 and still in service as of this writing. Three-stage solid rocket motor design: Stage 1 (Thiokol/ATK/NGIS, Promontory UT), Stage 2 (Aerojet, Sacramento CA), Stage 3 (Hercules/ATK/NGIS, Bacchus Works UT). Approximately 400 missiles remain on alert.

Peacekeeper (MX) The US Air Force's second-generation land-based ICBM, deployed 1986–2005. Four-stage solid motor design with stages from four different manufacturers (Thiokol, Aerojet, Hercules, Rocketdyne). Retired under START II Treaty terms; motors converted to Minotaur launch vehicles.

Pegasus An air-launched orbital rocket developed by Orbital Sciences Corporation with solid rocket stages built by Hercules/ATK. First flight 1990. Dropped from a carrier aircraft at altitude, then fires its three solid stages to reach orbit. The first privately developed orbital launch vehicle stages. Final operational flights occurring circa 2021–2026.

Polaris The US Navy's first submarine-launched ballistic missile, deployed 1961–1980. Development drove the large-scale industrialization of solid rocket propulsion. Motors supplied by Hercules (Bacchus Works) and Aerojet. Enabled the survivable second-strike capability that anchored Cold War deterrence strategy.

SLS (Space Launch System) NASA's heavy-lift launch vehicle using two five-segment solid rocket boosters (Northrop Grumman, Promontory UT) as the direct successor to the Space Shuttle SRBs. First flight: Artemis I, November 2022.

SLBM (Submarine-Launched Ballistic Missile) A ballistic missile designed to be launched from a submerged submarine. Current US SLBM: Trident II D5. SLBMs are the most survivable leg of the nuclear triad.

SRB (Solid Rocket Booster) A large solid rocket motor used to augment the thrust of a launch vehicle at liftoff. The Space Shuttle SRBs (Thiokol/ATK) and SLS SRBs (ATK/Northrop Grumman) are the largest solid rocket motors ever flown.

SRM (Solid Rocket Motor) General term for any rocket motor using solid propellant. Used interchangeably with "solid rocket" in this book.

Trident II D5 The current US Navy submarine-launched ballistic missile. Three solid stages, all manufactured by Northrop Grumman Innovation Systems (Hercules/ATK heritage, Bacchus Works). The primary sea-based nuclear deterrent for both the US and UK.

Organizational and Policy Terms

BRAC (Base Realignment and Closure) A statutory process by which Congress authorizes the Department of Defense to close or reorganize military installations. Multiple BRAC rounds in the 1990s (1991, 1993, 1995, 2005) eliminated government facilities that had provided sustaining work for the solid rocket industrial base.

FTC (Federal Trade Commission) The US regulatory body responsible for reviewing mergers and acquisitions for anticompetitive effects. The FTC has played a central structural role in the solid rocket industry — approving the Northrop/Orbital ATK merger with conditions (2018), blocking the Lockheed/Aerojet Rocketdyne merger (2022), and approving the L3Harris/Aerojet Rocketdyne merger (2023).

Nunn-McCurdy Act A federal law requiring the Department of Defense to notify Congress when a major defense program's unit cost grows beyond specified thresholds (15% = "significant"; 25% = "critical"). A critical breach requires the Secretary of Defense to certify that the program is essential to national security before it can continue. The LGM-35A Sentinel ICBM experienced a Nunn-McCurdy breach in 2024.

Sherman Antitrust Act (1890) The primary US federal antitrust statute, prohibiting monopolization and restraint of trade. The 1912 breakup of DuPont's explosives monopoly under the Sherman Act created Hercules Powder Company — the origin of the Hercules lineage that eventually became part of Northrop Grumman's solid rocket heritage.

Sole-Source Contract A government procurement contract awarded to a single supplier without competitive bidding, typically because only one contractor can provide the required capability. Sole-source relationships are common in strategic solid rocket motor production, where the technical requirements and cleared facilities limit competition to one or two qualified suppliers.